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Master's Thesis

用於三維合成孔徑超聲波成像之深度稀疏全晶片波束
形成架構

Depth-Sparse 3-D Synthetic Aperture Ultrasonic Imaging
with a Fully On-Chip Beamforming Architecture

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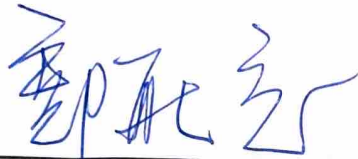
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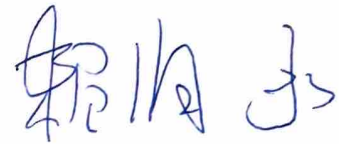
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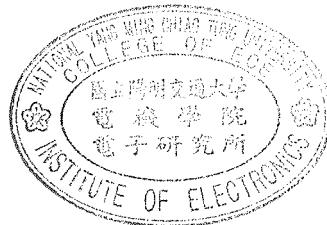
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摘 要

近年來，隨著自駕車、無人機、智慧工廠及智慧倉儲的快速發展，對於環境感知系統的需求日益增加。目前常見的環境感知技術包括雷達、光達以及可見光相機。然而，這些感測技術各自存在限制。雷達由於通道數較少，較難進一步提升空間解析度；光達雖具備高解析度，但成本較高且易受雨雪等天候條件影響；可見光相機則缺乏直接的深度資訊，限制其在部分應用場景中的可靠性。

為了解決上述問題，本論文提出一個三維超聲波合成孔徑環境偵測系統，並於 ZCU102 可程式化邏輯陣列上實現其硬體加速架構。所提出之系統利用合成孔徑技術提升傳統超聲波系統的角解析度，並透過硬體加速與演算法優化實現即時影像生成。此外，本研究進一步利用影像在深度方向上的稀疏特性，設計深度稀疏記憶體架構，有效降低合成孔徑成像所需的大量記憶體需求，使系統得以完全實現於晶片內。

實驗結果顯示，在視野角為正負 60 度、偵測距離為 0.3 至 2 公尺的系統設定下，所提出之系統於 ZCU102 平台上可達到 68.56 張每秒的影像生成速度。在角解析度方面，於 1.5 公尺處可達到最佳主瓣寬度為 2.86 度，顯示本系統能夠在維持高影像更新率的同時，有效提升環境感知的解析能力。

關鍵字：超聲波成像、合成孔徑、深度稀疏性、可程式化邏輯陣列、即時環境感知

Depth-Sparse 3-D Synthetic Aperture Ultrasonic Imaging with a Fully On-Chip Beamforming Architecture

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Abstract

In recent years, with the rapid development of autonomous vehicles, drones, smart factories, and smart warehouses, the demand for environment perception systems has significantly increased. Commonly used sensing technologies include radar, LiDAR, and visible-light cameras. However, these technologies have inherent limitations. Radar suffers from limited spatial resolution due to its small channel count. LiDAR provides high resolution but has high cost and is sensitive to adverse weather conditions such as rain and snow. Visible-light cameras lack direct depth information, which limits their reliability in certain scenarios.

To address these challenges, this thesis proposes a 3D synthetic aperture ultrasound environment sensing system and implements a hardware accelerator on a ZCU102 field-programmable gate array (FPGA) platform. The proposed system utilizes synthetic aperture techniques to improve the angular resolution of conventional ultrasound systems. Hardware acceleration and algorithm optimization are applied to achieve real-time imaging performance. Furthermore, depth sparsity in the imaging space is exploited to design a sparse memory architecture, which significantly reduces memory requirements and enables fully on-chip implementation.

Experimental results show that under a field of view of ± 60 degrees and a sensing range from 0.3 m to 2 m, the proposed system achieves a frame rate of 68.56 frames per second on the ZCU102 FPGA platform. In terms of angular resolution, the system achieves a minimum

mainlobe width of 3.06 degrees at a distance of 1.5 m. These results demonstrate that the proposed system can achieve high-resolution environment sensing while maintaining real-time performance.

Keywords: Ultrasound imaging, synthetic aperture, depth sparsity, FPGA, real-time environment sensing.



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Chapter 1. Introduction

In recent years, with the rapid development of autonomous vehicles, drones, and smart factories, environmental sensing systems have become essential for navigation, obstacle avoidance, and target localization. While common systems like radar, LiDAR, and cameras are widely used, they possess inherent limitations in cost, angular resolution, or sensitivity to light and weather conditions.

In comparison, ultrasound environmental sensing offers several advantages: it operates independently of light, is capable of sensing transparent objects, and is significantly more low-cost and scalable for indoor navigation and smart warehousing. However, as shown in Table 1, traditional ultrasonic systems are often relegated to simple proximity detection rather than high-resolution 3D imaging due to constraints in physical aperture size and channel count.

To enhance the resolution of ultrasonic systems, Synthetic Aperture (SA) techniques have been widely adopted. By leveraging multiple signal transmissions and receptions, this technique synthesizes measurement results to form an effective aperture equivalent to a larger physical one, thereby improving angular resolution and imaging quality.

However, implementing SA in ultrasonic environmental sensing systems introduces several critical challenges. First, SA imaging requires a large number of delay calculations and beamforming operations, which leads to high computational complexity. Second, multiple sub-images generated from different transmit events must be stored and accumulated, resulting in significant memory requirements. Finally, because a complete image frame can only be reconstructed after several transmit emissions, the frame rate is inherently limited, making real-time

imaging difficult to achieve.

To address these challenges, this thesis proposes a depth-sparsity-based 3D ultrasonic synthetic aperture imaging architecture implemented on a Field-Programmable Gate Array (FPGA). The proposed system utilizes synthetic aperture techniques to improve angular resolution beyond the limitations of conventional ultrasound systems. To reduce the computational complexity associated with delay calculation, a linear approximation method is employed. In addition, a dataflow pipeline architecture is designed to improve processing efficiency and maximize system throughput.

Furthermore, the proposed architecture exploits the sparsity along the depth dimension by storing only valid pixels instead of the complete image volume. This sparse memory scheme significantly reduces memory consumption and enables the beamforming process to be fully implemented within the programmable logic (PL) of a single FPGA. As a result, the dependency on external memory access is greatly reduced, improving overall system performance and making real-time 3D imaging feasible.

By employing the aforementioned methods, this thesis realizes a depth-sparsity-based 3D ultrasonic environmental sensing system on an FPGA platform, achieving both high resolution and real-time processing capability. The proposed architecture enhances imaging resolution while significantly reducing memory consumption and system latency through efficient hardware design. As a result, the system is well suited for real-time environmental sensing applications and provides a practical solution for fully on-chip ultrasonic imaging systems.

The remainder of this thesis is organized as follows. Chapter 2 reviews the background of synthetic aperture imaging, adaptive and coherence-based beamforming methods, 2-D sparse arrays, and real-time airborne 3-D ultrasound imaging systems. Chapter 3 presents the implementation details of the proposed depth-sparse 3-D synthetic aperture ultrasound imaging

system. Chapter 4 describes the system methodology, including the implementation and optimization techniques. Chapter 5 presents the experimental results and compares the imaging accuracy with the conventional delay-and-sum (DAS) beamforming method. Finally, Chapter 6 concludes the thesis and discusses directions for future work.



Chapter 2. Background and Motivation

2.1 Background

2.1.1 Linear Approximation for Delay Calculation

Accurate delay calculation is a critical component in ultrasound beamforming, as it directly affects focusing quality and spatial resolution. The propagation delay is typically determined by computing the round-trip distance between the imaging point and each transducer element, which involves computationally intensive square-root operations. To reduce hardware complexity and memory requirements, approximation methods have been proposed.

$$d = \sqrt{(x_S - x_D)^2 + (y_S - y_D)^2 + (z_S - z_D)^2} \quad (2.1)$$

Direct evaluation of this expression for every voxel-channel pair results in extremely high computational complexity, especially in volumetric imaging systems where billions of delay values must be generated per frame. To alleviate this bottleneck, we adopt the linear approximation strategy proposed in [1], where the delay difference relative to a reference scanline is approximated using a first-order Taylor expansion.

Following the derivation in [1], the square-root term can be expressed in the form $\sqrt{1+x}$ and approximated by its first-order expansion:

$$\sqrt{1+x} \approx 1 + \frac{1}{2}x \quad (2.2)$$

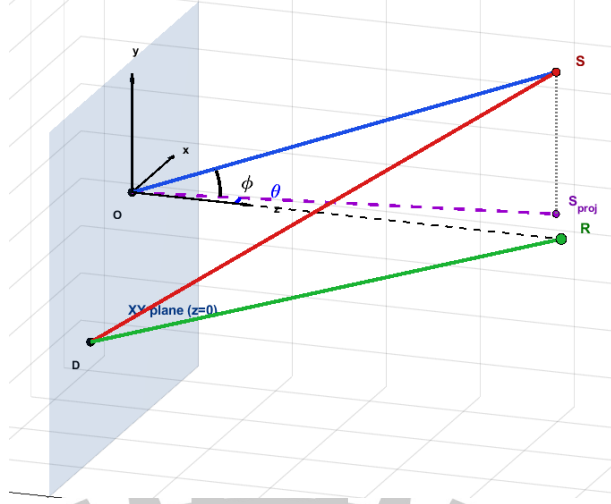


Figure 1: Coordinate geometry used for the delay approximation. O denotes the array center and D represents a receive element located on the transducer plane (XY plane). S is the target point in the imaging space. The steering direction toward S is parameterized by the azimuth angle θ and elevation angle ϕ . S_{proj} denotes the projection of S onto the XZ plane, and R represents the reference point along the steering direction. This coordinate representation is used to derive the linear approximation of the receive propagation distance.

Using this approximation, the delay correction term can be simplified into a linear function of the transducer element coordinates:

$$\Delta t \approx -\frac{x_D \sin \theta}{c} - \frac{y_D \sin \phi \cos \theta}{c} \quad (2.3)$$

This transformation eliminates the need for runtime square-root operations in the receive delay computation, reducing the delay generation to simple additions and multiplications. As discussed in [1], this first-order approximation provides sufficient accuracy for beamforming while significantly reducing hardware complexity and memory bandwidth requirements.

By adopting this linearized delay model, the proposed architecture achieves a favorable trade-off between computational efficiency and focusing accuracy, enabling fully on-chip implementation for real-time 3-D synthetic aperture imaging.

2.1.2 Adaptive and Coherence-Based Methods

In conventional delay-and-sum (DAS) beamforming, all receive channels are equally weighted during image reconstruction. This uniform weighting scheme makes DAS sensitive to noise and often results in poor contrast and elevated sidelobe levels, especially under low signal-to-noise ratio (SNR) conditions. To overcome this limitation, adaptive and coherence-based beamforming methods have been developed to improve focusing quality and suppress incoherent interference.

Among these approaches, the coherence factor (CF), originally proposed by Li and Li[2], has become one of the most widely adopted techniques. The CF measures the degree of coherence among signals received across the aperture and applies it as an adaptive weighting to the beamformed output. By emphasizing coherent signal components while attenuating incoherent contributions, CF effectively reduces sidelobe artifacts and speckle noise, resulting in significant contrast enhancement.

Due to its computational simplicity and effectiveness, the CF method has been extensively applied in both medical and air-coupled ultrasound imaging systems. It provides an efficient means to improve image quality under low-SNR conditions and has become a fundamental component in modern beamforming architectures, particularly in embedded and real-time imaging implementations.

2.1.3 Synthetic Aperture

Synthetic aperture (SA) is a well-established technique for enhancing image quality and spatial resolution. It has been widely adopted in radar systems, where platform motion, such as that of satellites or aircraft, is utilized to synthesize a large effective aperture, thereby achieving fine angular resolution beyond the physical limitations of the antenna size. By coherently

integrating echoes acquired at multiple spatial positions, SA systems can significantly improve target separability and imaging accuracy.

In contrast, ultrasonic imaging systems are typically stationary or exhibit non-deterministic motion, requiring a different implementation strategy. Instead of relying on platform movement, ultrasonic synthetic aperture imaging employs a fixed array of transmitters and receivers, and the effective aperture is synthesized by coherently combining signals obtained from multiple transmit–receive events. This approach enables the system to achieve improved focusing quality and spatial resolution without increasing the physical aperture size.

Synthetic aperture ultrasound imaging has been extensively studied and demonstrated to significantly enhance lateral resolution compared to conventional delay-and-sum (DAS) beam-forming methods. Jensen et al. [3] showed that SA imaging can approach the theoretical resolution limit defined by the aperture size. Similarly, Park and Khuri-Yakub [4] demonstrated 3-D airborne ultrasound synthetic aperture imaging using CMUT arrays, achieving high-resolution volumetric imaging by exploiting multi-element synthetic aperture reconstruction.

Furthermore, the resolution improvement provided by synthetic aperture techniques is particularly important in air-coupled ultrasound imaging, where the limited number of transducer elements and low signal-to-noise ratio often restrict imaging performance. Recent studies have shown that synthetic aperture imaging can effectively reduce main-lobe width and improve target resolution, making it a key technique for high-resolution airborne ultrasound imaging systems [5, 6].

2.1.4 2-D Sparse Array

The 2-D sparse array adopted in this work follows the Fermat spiral (sunflower) geometry, which has been theoretically and experimentally validated as an effective sparse sampling

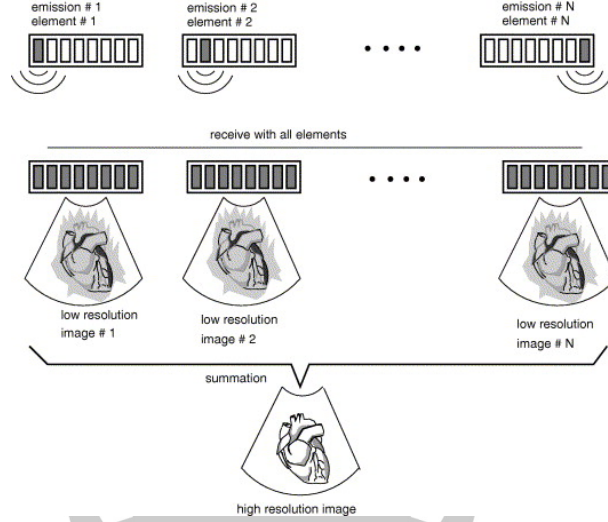


Figure 2: Basic principle of synthetic aperture (SA) ultrasound imaging. Each transmit event generates a low-resolution image using one active transmitter and all receive elements. The final image is reconstructed by coherently combining the responses from multiple transmit events, effectively synthesizing a larger aperture and improving spatial resolution.

strategy for acoustic imaging. Sarradj [7] systematically compared random, uniform, and spiral microphone configurations, demonstrating that the sunflower spiral provides an optimal trade-off between main-lobe width and sidelobe level due to its quasi-uniform angular distribution and radial density balance.

By rotating each element by the golden angle (approximately 137.5°), the array achieves non-redundant spatial sampling while maintaining radial symmetry. This property minimizes grating lobes and preserves isotropic beam characteristics even under significant channel reduction. Compared with a fully populated array, the sunflower sparse configuration maintains comparable FWHM performance while significantly reducing hardware complexity.

Furthermore, the slow speed of sound in air (approximately 340 m/s) introduces additional temporal constraints, as each transmission must wait for echo decay before subsequent firing. This makes multi-transmit compounding less efficient than in medical ultrasound, increasing the importance of computationally efficient receive beamforming and sparse array strategies.

Therefore, achieving real-time 3-D airborne ultrasonic imaging requires a careful balance among sparse array geometry, hardware-efficient beamforming architecture, and memory op-

timization. The proposed system addresses these challenges by integrating a sunflower sparse array, and fully streaming FPGA acceleration to enable near real-time 3-D imaging under strict hardware constraints.

2.2 Related Work and Motivation

2.2.1 Disadvantage of Typical Environmental Sensing System

In recent years, with the rapid development of autonomous vehicles, drones, smart warehouses, and smart factories, environmental sensing systems have become essential to various autonomous systems. These systems must be able to actively sense the environment for navigation, obstacle avoidance, and target localization. Common environmental sensing systems include radar, LiDAR, and camera systems. However, these systems have inherent limitations. Radar systems exhibit superior environmental adaptability; Nevertheless, their angular resolution is difficult to improve further due to the limited number of channels. As for LiDAR systems, although they can provide high-resolution environmental information, they are expensive and easily affected by weather conditions. Last but not least, camera systems suffer from a lack of depth information and are heavily influenced by different light conditions [8, 9].

Table 1: Comparison of camera, LiDAR, and RADAR for environment sensing

Sensor	Detection Dimension	Cost	Accuracy	Robustness
Camera	2D	Low	Limited depth information	Sensitive to light and weather
LiDAR	3D	High	High	Sensitive to weather
RADAR	3D	Medium	Limited angular resolution	High

In comparison, ultrasound environmental sensing system has several advantages. First, it can operate independent of light conditions. Furthermore, ultrasound environmental sensing system shows more robustness under different weather conditions. Therefore, it is particularly well-suited for applications in close range sensing for autonomous vehicles and drones. What's

more, the ultrasound environmental sensing system is low-cost, more scalable and capable of sensing transparent object, making it an ideal solution for indoor navigation, smart warehousing and industrial environment. Nevertheless, the resolution of conventional ultrasonic systems is constrained by the physical aperture size and the number of channels, limiting their efficacy in high-resolution 3D imaging applications.

2.2.2 Low-SNR Challenge in Air-Coupled Ultrasound Imaging

Compared with conventional medical ultrasound imaging performed in water or biological tissues, air-coupled ultrasonic imaging suffers from a significantly lower signal-to-noise ratio (SNR). The primary reason is the large acoustic impedance mismatch between air and solid materials, which severely limits the amount of acoustic energy that can be transmitted and received. The acoustic impedance mismatch between the transducer [10] [11], the inspected object, and the surrounding air introduces substantial attenuation. Previous studies have reported that conventional immersion transducers may experience amplitude losses of up to 90 dB when used in air-coupled configurations, while the test specimen itself may introduce additional attenuation on the order of 70 – 80 dB [12]. As a consequence, only a small portion of the transmitted acoustic energy can be detected by the receiving transducers, leading to weak echo signals and degraded imaging performance. Under such low-SNR conditions, conventional delay-and-sum (DAS) beamforming often produces images with elevated sidelobes and poor contrast. Synthetic aperture (SA) imaging provides an effective solution to this limitation. By coherently integrating echoes from multiple transmit–receive events, SA increases the effective aperture and improves signal coherence, thereby enhancing spatial resolution and robustness in low-SNR airborne ultrasound environments.

2.2.3 Ultrasonic Imaging Systems

Medical ultrasound imaging has been widely adopted in clinical diagnosis because it is safe, noninvasive, portable, relatively low-cost, and capable of real-time imaging. These characteristics make ultrasound particularly suitable for point-of-care (POC) applications, where portability and immediate imaging feedback are highly desirable. As portable and bedside imaging systems become increasingly important, substantial research efforts have focused on implementing ultrasound signal processing and imaging pipelines on compact and power-efficient hardware platforms.

In ultrasound imaging systems, receive beamforming is a critical signal processing step that significantly influences image quality. Proper beamforming improves signal-to-noise ratio (SNR), spatial resolution, and contrast resolution. However, conventional dynamic receive beamforming requires delay updates for every focal point and every receive channel, resulting in considerable computational complexity and memory access overhead.

Several studies have investigated efficient hardware architectures for portable ultrasound imaging systems. Kim *et al.* [13] proposed a single FPGA-based portable ultrasound imaging system for point-of-care applications. In their design, major ultrasound signal and image processing modules were integrated into a single FPGA device. The system adopted pseudo-dynamic receive focusing and lookup-table (LUT) based delay generation to reduce hardware complexity while still achieving real-time B-mode imaging. This work demonstrated the feasibility of implementing compact ultrasound imaging systems using FPGA-based architectures.

To further reduce beamforming complexity, Yoon *et al.* [14] developed a low-complexity digital beamformer architecture for high-frequency ultrasound imaging. Their approach generated receive delays using a LUT combined with linear interpolation between predetermined focal points. In addition, a postfiltering technique was introduced to group channel data with iden-

tical fractional delays before interpolation, thereby reducing the number of required polyphase filters. Their results showed that the proposed architecture maintained lateral and axial resolution while significantly lowering computational complexity.

For ultra-compact ultrasound imaging systems, Kang *et al.* [15] proposed an efficient parallel beamforming architecture based on a shared FIFO structure. Their design addressed the challenge of generating multiple scanlines simultaneously without incurring excessive hardware cost. By introducing a shared FIFO structure on top of a post-fractional delay filtering architecture, the system achieved beamforming accuracy comparable to conventional parallel beamforming while reducing power consumption.

2.2.4 FPGA-Based Beamforming Architectures

With the increasing demand for real-time and high-resolution ultrasound imaging, many researchers have explored hardware acceleration using field-programmable gate arrays (FPGAs). FPGA platforms provide massive parallelism and pipelined processing capability, making them well suited for computationally intensive beamforming operations.

Sampson *et al.* [16] presented an FPGA implementation of a low-power three-dimensional ultrasound beamformer based on the Sonic Millip3De architecture. Their design demonstrated the feasibility of real-time 3-D synthetic aperture beamforming under stringent power constraints. To reduce the high computational cost associated with exact delay calculation and large lookup tables, the authors proposed an iterative delay estimation method using simple arithmetic operations and a compact coefficient table.

More recently, Kou and Oelze [17] investigated FPGA-based beamforming architectures for ultrafast wearable ultrasound imaging. Their design migrated beamforming inputs from RF data to IQ data, significantly reducing sampling rate requirements, data transfer bandwidth,

and overall power consumption. Implemented using Vitis HLS on a Xilinx KR260 platform, their work demonstrated the feasibility of low-power, high-frame-rate ultrasound imaging using FPGA-based architectures.

In addition to conventional beamforming architectures, synthetic aperture imaging has also been implemented on FPGA platforms to improve spatial resolution. Amaro *et al.* [18] presented a high-level synthesis (HLS) framework for FPGA beamformers designed for synthetic aperture imaging. Their results showed that SA beamforming can be efficiently implemented through parallel processing and pipeline architectures.

Similarly, Vogel *et al.* [19] proposed a scalable single-FPGA architecture capable of performing real-time three-dimensional ultrasound beamforming. Their design employed a highly parallel architecture that supports volumetric imaging while maintaining low power consumption and high image quality.

Ricci *et al.* [20] further proposed a multi-cycle parallel FPGA architecture for real-time ultrasound signal processing. By exploiting parallelism and pipelined processing, their design significantly accelerated beamforming operations and enabled real-time processing for ultrasound systems with a large number of channels.

These studies demonstrate that FPGA-based architectures are an effective solution for supporting real-time ultrasound imaging with high spatial resolution. However, most existing works focus on medical ultrasound imaging in biological tissue. In contrast, this thesis targets three-dimensional air-coupled ultrasonic synthetic aperture imaging, where lower acoustic velocity, weaker echo signals, and more severe SNR limitations introduce additional challenges in memory architecture design, delay computation, and real-time processing.

2.2.5 Synthetic Aperture Imaging for Resolution Enhancement

Conventional delay-and-sum (DAS) beamforming suffers from limited spatial resolution due to the restricted physical aperture size and channel count [3], [21]. Synthetic aperture (SA) imaging has been widely proposed to overcome this limitation by coherently combining multiple transmit–receive events to synthesize a larger effective aperture, thereby significantly improving angular resolution [18]. Park and Khuri-Yakub experimentally showed that the resolution of airborne 3-D ultrasound imaging is fundamentally determined by the effective aperture size and bandwidth, achieving a 6 dB FWHM of approximately 2.7° [4].

Similarly, Allevato et al. reported an angular resolution of approximately 12° using an 8E8 air-coupled phased array, highlighting the resolution limitations imposed by small apertures [5]. Adaptive beamforming techniques such as minimum variance (MV) beamforming have also been proposed to further improve resolution beyond DAS, at the cost of significantly increased computational complexity [21, 22]. These studies confirm that achieving high spatial resolution remains a fundamental challenge in real-time airborne ultrasound imaging systems.

Therefore, improving angular resolution while maintaining real-time performance and hardware efficiency remains a key research challenge in synthetic aperture ultrasound imaging.

Chapter 3. System Architecture

3.1 System Overview

Fig. 3 illustrates the overall architecture of the proposed depth-sparse 3-D synthetic aperture ultrasound imaging system. The design follows a fully streaming dataflow paradigm, where memory index for each channel is generated concurrently and sent into input buffer to get the RF data channel-by-channel and propagated through beamforming, and sparse accumulation units without external memory dependency.

The system begins with an AXI-Stream interface that feeds the Input Buffer. This module temporarily stores multi-channel RF data and selects the data based on the index from delay calculation stage, distributing them to the subsequent beamforming stages.

The Delay Calculator generates precise propagation delays for each channel using both transmit (TX) and receive (RX) distance computation. A hybrid approach combining distance calculation and lookup tables (including channel geometry, sine/cosine tables, and distance from central scan point to each receive (T_ord)) is employed to balance accuracy and hardware efficiency. The delay generator provides per-channel delay indices to support dynamic receive focusing in synthetic aperture imaging.

In the Beamformer stage, each channel first undergoes quadrature demodulation. The delayed samples are then aligned and coherently summed across channels. A coherence-factor (CF) scaling unit is integrated to enhance image contrast and suppress incoherent noise, improving spatial resolution under low-SNR air-coupled conditions.

The Vector Adder and Sparse Memory Unit perform partial image accumulation using an adder tree structure. Instead of storing the full 3-D voxel grid, only valid depth points are maintained in a sparse memory buffer. A Top-K buffer and sparse memory update unit dynamically manage significant voxel responses, enabling depth-sparsity exploitation and reducing on-chip memory requirements.

Finally, the processed 3D image voxels are streamed out via AXI-Stream. The entire architecture is designed to achieve fully on-chip implementation, minimizing off-chip DRAM bandwidth and enabling real-time 3-D airborne ultrasound imaging.

The imaging volume spans a $\pm 60^\circ$ angular field in both azimuth and elevation, with a depth range of 0.3–2 m. The angular domain is discretized into 69 sampling steps per dimension, yielding 69×69 beam directions. Each beam direction contains 284 depth samples, forming a volumetric grid of $69 \times 69 \times 284$ voxels.

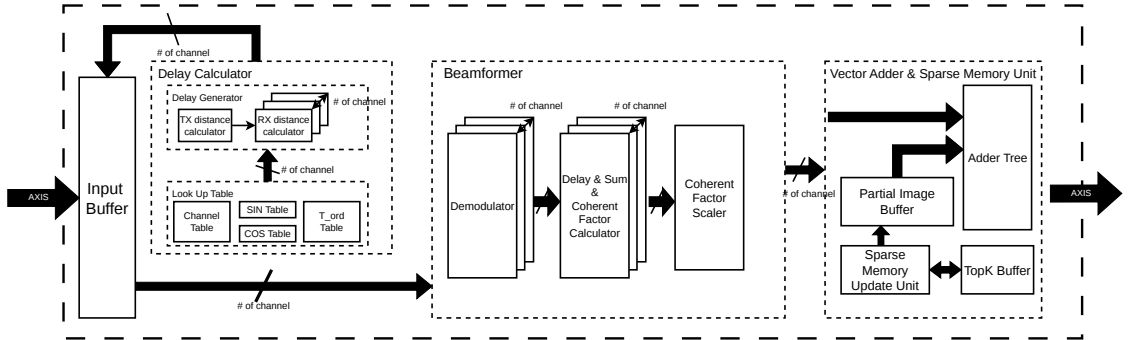


Figure 3: Overall architecture of the proposed depth-sparse 3-D synthetic aperture ultrasound imaging system. The system adopts a fully streaming dataflow architecture composed of three main modules: the delay calculator, the beamformer, and the vector adder with sparse memory update unit. The delay calculator generates per-channel memory indices, the beamformer performs quadrature demodulation and delay-and-sum beamforming with coherence factor scaling, and the vector adder accumulates partial images while maintaining a depth-sparse voxel buffer.

3.1.1 Pipeline Hierarchy

To achieve real-time performance, the proposed architecture adopts a hierarchical pipeline design consisting of both coarse-grain dataflow pipelining and fine-grain computation pipelin-

ing. This hierarchy enables high throughput while maintaining manageable hardware complexity.

Fig. 4 illustrates the coarse-grain dataflow pipeline across major system modules. In a naive sequential implementation, each processing stage—including delay calculation, beamforming, and vector accumulation—would execute one after another. As a result, the total event latency would be the sum of all stages.

To improve throughput, the architecture employs a dataflow pipeline where these processing stages operate concurrently. While one scanline is being processed by the beamformer, the delay calculator can simultaneously generate the delay indices for the next scanline, and the vector adder can accumulate results from the previous scanline. This overlapping execution significantly improves the system throughput, as multiple scanlines can be processed simultaneously in different pipeline stages.

In addition to the coarse-grain dataflow pipeline, the architecture further incorporates a fine-grain pipeline within each processing stage, as illustrated in Fig. 5. Instead of waiting for the entire scanline to complete, the system processes data at the pixel level. Once the delay of pixel i is computed, the corresponding beamforming operation can immediately begin while the delay calculator proceeds to pixel $i + 1$. Similarly, the vector accumulation stage starts processing pixel i as soon as its beamformed value becomes available.

This fine-grain pipelining allows the system to achieve an initiation interval (II) of one cycle, meaning a new voxel can enter the pipeline every clock cycle. Consequently, multiple voxels are processed concurrently at different stages of the pipeline, dramatically improving computational throughput.

By combining coarse-grain module-level dataflow with fine-grain pixel-level pipelining, the proposed architecture maximizes hardware utilization and sustains the high processing through-

put required for real-time 3-D synthetic aperture ultrasound imaging.

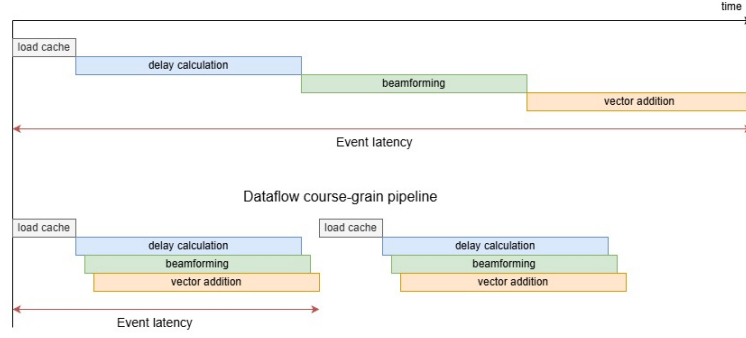


Figure 4: Illustration of the coarse-grain dataflow pipeline in the proposed architecture. Without pipelining, delay calculation, beamforming, and vector addition are executed sequentially, resulting in long event latency. By applying a dataflow pipeline, these stages operate concurrently for different data blocks, allowing computation stages to overlap and significantly improving throughput.

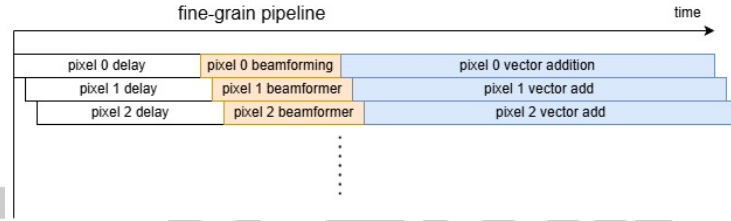


Figure 5: Fine-grain pipeline across processing stages. Each pixel sequentially passes through delay calculation, beamforming, and vector addition. By pipelining these operations, multiple pixels are processed concurrently in different stages, allowing the architecture to achieve an initiation interval of one pixel per clock cycle after pipeline filling.

3.2 Channel-level Parallelism

Fig. 6 illustrates the channel-level parallelism adopted in the proposed beamforming architecture. In implementation by MCU, samples from different receive channels are processed sequentially, which significantly limits throughput when a large number of channels are involved.

To overcome this limitation, the proposed design processes all receive channels in parallel. The delay calculator first generates a delay index for each receive channel simultaneously. These indices are then forwarded to the input buffer, which retrieves the corresponding RF

samples from memory. As a result, a set of delayed samples corresponding to all channels is produced concurrently.

The input buffer outputs one delayed sample per channel in each cycle, denoted as $\text{sample}_0, \text{sample}_1, \dots$, where N_{ch} represents the total number of receive channels. These samples are then streamed directly into the beamformer stage.

Inside the beamformer, all channel samples are processed simultaneously through parallel demodulation and delay-and-sum operations. This channel-level parallelism allows the system to perform coherent summation across all channels within a single pipeline stage, greatly improving computational throughput.

By exploiting channel-level parallelism, the proposed architecture eliminates the need for iterative channel processing and enables the beamformer to sustain the high data rate required by the fine-grain pipeline. Combined with the dataflow pipeline structure, the system can continuously process incoming voxels while maintaining an initiation interval (II) of one cycle.

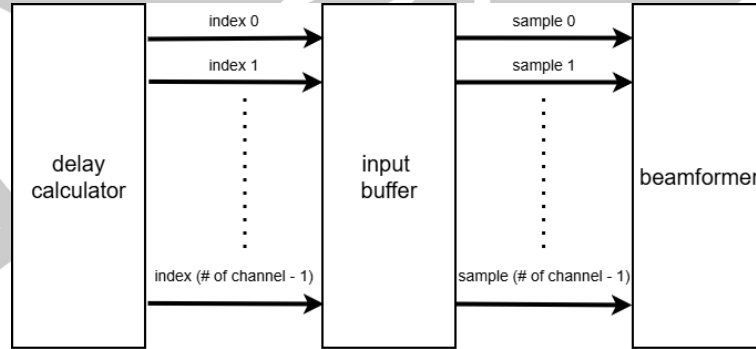
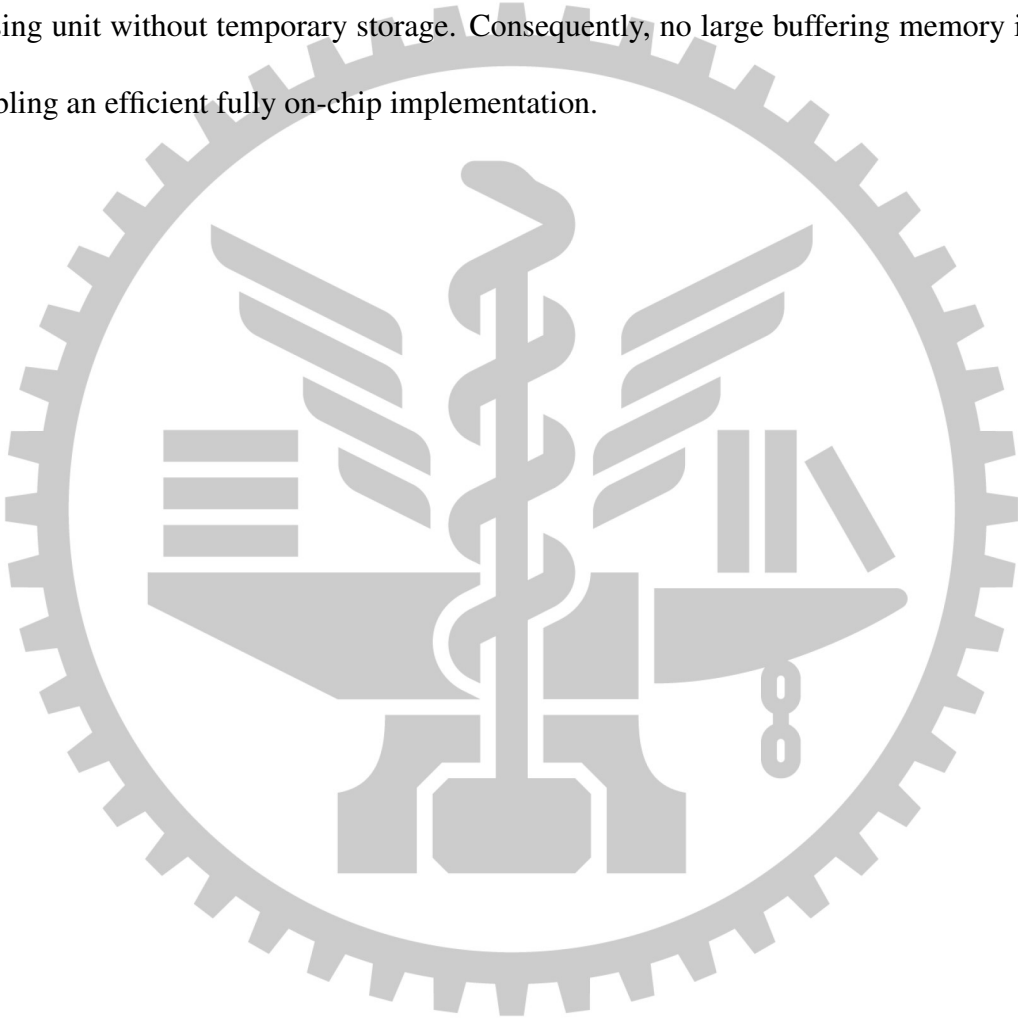


Figure 6: Channel-level parallelism in the proposed architecture. The delay calculator generates memory indices for all receive channels concurrently. These indices are used by the input buffer to fetch the corresponding RF samples, which are then streamed to the beamformer for parallel processing across channels.

3.3 Delay Calculator

Fig. 7 presents the architecture of the proposed delay calculator. This stage constitutes the computationally intensive component of the system, as it executes real-time delay generation.

However, memory consumption remains extremely low. The architecture adopts a fully pipelined streaming dataflow, where intermediate results are forwarded directly to the next processing unit without temporary storage. Consequently, no large buffering memory is required, enabling an efficient fully on-chip implementation.



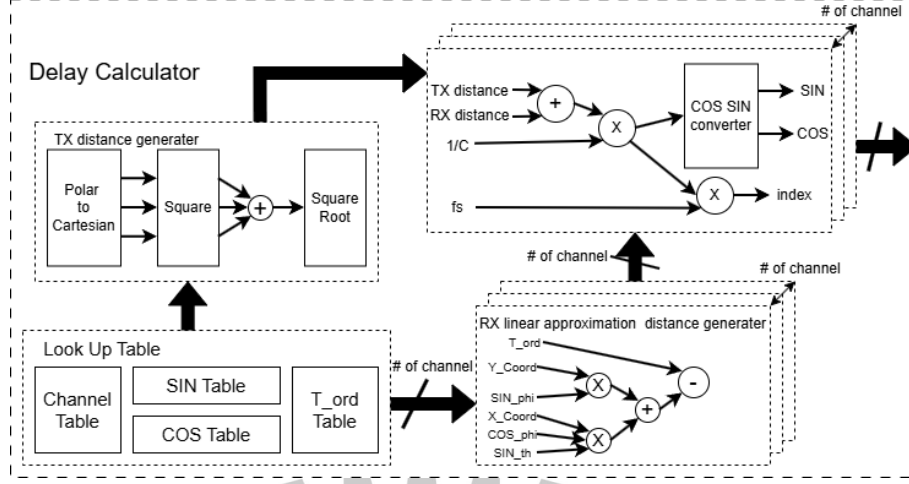


Figure 7: Proposed delay calculator architecture. The transmit propagation distance is obtained through direct geometric computation, while the receive distance is approximated using a linear delay model. Lookup tables storing channel coordinates, sine/cosine values, and T_{rd} are used to reduce hardware complexity. The computed delays are converted into memory indices for each receive channel to support parallel beamforming.

3.3.1 Linear Approximation of Receive Delay

In the proposed system, T_{ord} represents the reference propagation distance from the array origin to a point located at depth r along the normal direction of the array plane, and subsequently to the D -th receive element. For an element located at $(x_D, y_D, 0)$, the exact reference distance is

$$T_{ord,D}(r) = \sqrt{r^2 + x_D^2 + y_D^2}. \quad (3.1)$$

For practical imaging ranges where $r \gg \sqrt{x_D^2 + y_D^2}$, a first-order Taylor expansion yields

$$T_{ord,D}(r) \approx r + \frac{x_D^2 + y_D^2}{2r}. \quad (3.2)$$

This term is precomputed and stored in a LUT, eliminating the need for runtime square-root operations.

Exact receive distance. For the D -th receive element located at $\mathbf{p}_D = [x_D, y_D, 0]^T$, the exact propagation distance is

$$d_D(r) = \|r\hat{\mathbf{u}}(\theta, \phi) - \mathbf{p}_D\|. \quad (3.3)$$

Expanding the squared distance yields

$$d_D^2(r) = r^2 - 2r\hat{\mathbf{u}}^T \mathbf{p}_D + \|\mathbf{p}_D\|^2. \quad (3.4)$$

Factoring out r^2 :

$$d_D(r) = r\sqrt{1 - \frac{2}{r}\hat{\mathbf{u}}^T \mathbf{p}_D + \frac{\|\mathbf{p}_D\|^2}{r^2}}. \quad (3.5)$$

First-order approximation. For practical imaging ranges ($r \gg \|\mathbf{p}_D\|$), the term inside the square-root is close to unity. Applying the first-order Taylor expansion

$$\sqrt{1+x} \approx 1 + \frac{1}{2}x, \quad (3.6)$$

we obtain

$$d_D(r) \approx r - \hat{\mathbf{u}}^T \mathbf{p}_D. \quad (3.7)$$

Therefore, the receive distance can be expressed as

$$d_D(r) \approx T_{\text{ord}}(r) + \Delta d_D, \quad (3.8)$$

where

$$\Delta d_D \approx -\hat{\mathbf{u}}^T \mathbf{p}_D. \quad (3.9)$$

Polar form. Using the adopted angle definition (Fig. 1), the projection term becomes

$$\hat{\mathbf{u}}^T \mathbf{p}_D = x_D \sin \theta + y_D \sin \phi \cos \theta, \quad (3.10)$$

leading to the implemented delay correction

$$\Delta t_D \approx -\frac{x_D \sin \theta}{c} - \frac{y_D \sin \phi \cos \theta}{c}. \quad (3.11)$$

Hence, the total receive delay is obtained as

$$t_D(r, \theta, \phi) = \frac{T_{\text{ord}}(r)}{c} + \Delta t_D. \quad (3.12)$$

This formulation eliminates the runtime square-root operation and reduces delay generation to simple multiplications and additions, making it suitable for fully streaming FPGA implementation.

Modification from T_{ord} to T_{rd} . In our geometry (Fig. 7), let O denote the array origin and R be the reference point located along the array-normal direction at range R from O . In the original formulation, the reference distance T_{ord} includes two segments: (i) the common path from the origin to the reference point ($|OR|$), and (ii) the element-dependent path from the reference point to the D -th receive element ($|RD|$). Therefore, for an element located at $(x_D, y_D, 0)$,

$$T_{\text{ord},D}(R) = |OR| + |RD| = R + \sqrt{x_D^2 + y_D^2 + R^2}. \quad (3.13)$$

Since the term $|OR| = R$ is identical for all channels, it does not affect relative delay differences across receive elements. To simplify the hardware implementation, we remove this common offset and define a reduced reference distance

$$T_{rd,D}(R) \triangleq T_{ord,D}(R) - |OR| = \sqrt{x_D^2 + y_D^2 + R^2}. \quad (3.14)$$

This modification is introduced due to the uncertainty of the transmitter (TX) position in the synthetic aperture configuration. The proposed system employs five emitters, and only one is located at the geometric center of the array. Therefore, the TX position cannot be assumed to coincide with the array origin for all transmission events.

As a result, the linear approximation strategy is applied only to the receive (RX) distance, where per-channel geometric variations dominate the computational complexity. In contrast, the transmit (TX) distance must be computed with respect to the actual active emitter location. Since the TX delay is identical for all receive channels within a given transmission, it is calculated exactly using a CORDIC-based square-root implementation.

Consequently, the final delay model adopts a hybrid approach: the TX propagation distance is evaluated exactly, while the RX propagation distance is approximated using first-order linearization. This separation ensures computational efficiency without sacrificing focusing accuracy in the multi-TX synthetic aperture framework.

3.3.2 Centralized Look Up Table

In the proposed delay calculator, the lookup tables are carefully organized to balance memory usage and computational complexity. The channel table stores the spatial coordinates of all receive elements, which are reused for every scanline. The sine and cosine tables contain the precomputed steering angles for each scanline direction, avoiding runtime trigonometric

evaluations.

The T_{rd} table stores the reference receive distance for every depth sample and receive channel. Since the receive delay varies across both depth and channel dimensions, this table requires 284×64 entries and therefore dominates the overall LUT memory consumption. The bitwidth of each entry is set to be 21 bits.

By precomputing these parameters, the delay calculator avoids expensive operations such as trigonometric functions and repeated geometric distance calculations during runtime. As a result, the delay computation is reduced to a sequence of additions and multiplications, which can be efficiently mapped onto the FPGA datapath and fully pipelined.

3.3.3 Delay Calculator Hardware Description

The proposed delay calculator is designed to generate memory indices for all channels concurrently. Through a fully pipelined architecture combined with channel-level parallelism, the hardware is capable of producing one complete set of memory indices per clock cycle, thereby sustaining a throughput sufficient for real-time beamforming.

As shown in Fig. 7, the delay calculator decomposes the propagation delay into transmit (TX) and receive (RX) components. The transmit distance is computed using the exact geometric distance calculation, which involves a square-root operation. Since the TX distance is identical for all channels in a given transmission event, it is computed once and shared across channels. In contrast, the receive delay varies across different receiver elements and therefore requires per-channel computation. To reduce hardware complexity, a linear approximation model is adopted for the RX delay calculation.

The delay indices for all receive channels are generated simultaneously. Each channel is assigned an independent computation path that evaluates the corresponding delay correction

based on the receiver coordinates and steering angles. This fully parallel channel architecture eliminates the need for iterative channel processing and allows the delay generator to produce all channel indices in a single cycle.

Furthermore, the delay calculator is implemented as a streaming datapath without feedback dependencies. Once the pipeline is filled, the architecture can continuously output delay indices at every clock cycle, achieving an initiation interval (II) of one. This design enables seamless integration with the downstream beamforming pipeline and supports the high throughput required for real-time 3-D synthetic aperture imaging.



3.4 Beamformer

Fig.8 illustrates the architecture of the proposed streaming beamformer. The input RF data are first buffered through a dataflow FIFO and fed into a quadrature demodulator. Each channel is multiplied by pre-generated cosine and sine references to obtain the in-phase (I) and quadrature (Q) baseband components.

After quadrature demodulation, the beamformer datapath is divided into two parallel computation branches. The first branch performs coherent summation across all channels using a DAS adder tree, followed by magnitude computation of the beamformed signal. The second branch computes the magnitude of each channel individually and accumulates them through a CF adder tree to obtain the incoherent energy term required for coherence factor evaluation.

These two branches operate concurrently in the streaming pipeline. Their outputs are later combined to compute the coherence factor, and the beamformed magnitude is multiplied by the CF value to produce the final voxel output. This parallel datapath allows the DAS and CF computations to proceed simultaneously while maintaining the fully pipelined streaming architecture.

The entire architecture is fully pipelined and streaming-based, enabling one pixel output per clock cycle after pipeline filling.

3.4.1 Coherent Factor

The most computationally demanding component of the beamformer is the coherence factor (CF) calculation. As shown in Fig. 8, the CF unit consists of per-channel squaring operations followed by a hierarchical adder tree to accumulate the signal energy across all channels.

The coherence factor evaluates the spatial coherence of the received aperture data and is

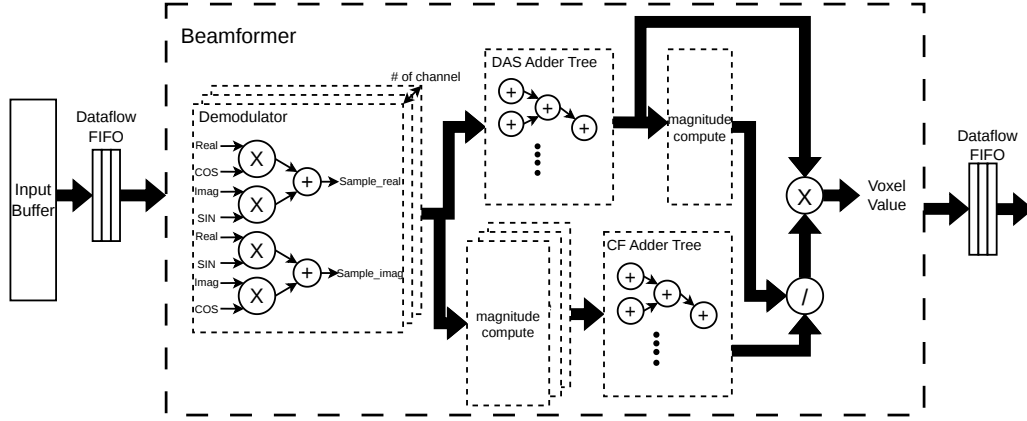


Figure 8: Architecture of the proposed streaming beamformer. The input RF samples are first demodulated into in-phase (I) and quadrature (Q) components using pre-generated sine and cosine references. The demodulated signals are coherently summed across channels using a delay-and-sum (DAS) adder tree. In parallel, the signal magnitudes are accumulated through a coherence factor (CF) adder tree to evaluate the spatial coherence of the received signals. The final voxel value is obtained by weighting the beamformed magnitude with the computed CF term.

defined as

$$CF = \frac{\left| \sum_{i=1}^N s_i \right|^2}{N \sum_{i=1}^N |s_i|^2}, \quad (3.15)$$

where s_i denotes the complex baseband signal of the i -th receive channel after delay compensation, and N represents the total number of channels. The numerator corresponds to the squared magnitude of the coherent summation, while the denominator represents the accumulated incoherent energy of all channels.

In hardware implementation, the denominator requires per-channel squaring operations (i.e., $\Re\{s_i\}^2 + \Im\{s_i\}^2$) followed by a reduction adder tree, which introduces additional multipliers and accumulation stages compared to the conventional delay-and-sum (DAS) operation. As a result, the CF computation consumes more DSP and logic resources than the DAS adder tree.

Nevertheless, incorporating the CF significantly enhances image contrast by penalizing incoherent signal components and suppressing sidelobe artifacts. When the received signals are highly phase-aligned, the coherent summation dominates and CF approaches unity. In contrast,

for incoherent noise or off-axis reflections, the coherent sum diminishes while the accumulated energy remains large, leading to a reduced CF value. This adaptive weighting mechanism substantially improves overall image quality under low-SNR air-coupled conditions.

The quantitative performance improvements introduced by the CF module will be demonstrated in the subsequent chapter. Therefore, the inclusion of the CF unit represents a justified trade-off between computational complexity and imaging performance.

3.4.2 Dataflow FIFO and Shift-Register Implementation

To sustain a fully pipelined architecture with an initiation interval (II) of one, the dataflow FIFOs inserted before and after the beamformer are implemented as shift-register-based streaming buffers.

Since the entire beamforming datapath is designed to generate one output sample per clock cycle after pipeline filling, no large-depth buffering is required between functional modules. Instead, each FIFO stage only needs to hold a small number of samples to maintain timing decoupling and data alignment. Therefore, the FIFOs are synthesized as distributed shift registers rather than BRAM-based memories.

This shift-register implementation offers several advantages. First, it minimizes memory resource usage by avoiding block RAM allocation. Second, it preserves the $II = 1$ throughput of the pipeline without introducing additional latency. Third, it naturally supports continuous streaming operation, which is essential for real-time 3-D beamforming.

Consequently, both the input and output dataflow FIFOs function as lightweight pipeline registers that propagate data forward at every clock cycle, ensuring deterministic timing and stable high-throughput operation.

3.5 Vector Adder and Memory Update Unit

Fig. 9 illustrates the module responsible for synthetic aperture accumulation. This component consumes the largest portion of on-chip memory, since five intermediate 3-D images corresponding to different transmit (TX) events must be stored.

The vector adder receives the beamformer output from the current transmission and accumulates it with the partial results generated by the other four TX events. Through this process, the final synthetic aperture response is formed by coherently combining the contributions from all transmitters.

Meanwhile, the Top-K update arbitrator and the temporal buffer, implemented using registers, maintain the indices and values of valid voxels. This mechanism enables efficient management of sparse voxel data and supports the depth-sparse accumulation strategy adopted in the proposed architecture.

This mechanism enables efficient management of sparse voxel data, ensuring that only significant voxels are preserved while avoiding the need to store the full volumetric image.

3.5.1 Sparse Memory for Synthetic Aperture

The 3-D image in the proposed system consists of $69 \times 69 \times 284$ voxels. For each voxel, both the real and imaginary components of the beamformed signal must be stored. Each component is represented using a 12-bit fixed-point format.

Consequently, storing the entire 3-D image on-chip would require a total memory size of 185,107,680 bits, which exceeds the on-chip memory capacity of most FPGA or ASIC implementations.

To address this limitation, a depth-sparse memory architecture is adopted. The Top-K up-

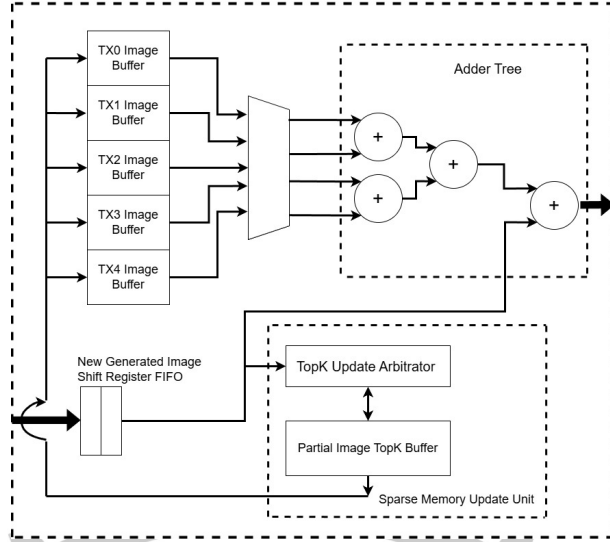


Figure 9: Architecture of the vector adder and sparse memory update unit. Partial images generated from five transmit events are stored in dedicated TX image buffers. The voxel values from these buffers are selected and coherently accumulated through a pipelined adder tree to form the synthetic aperture response. Meanwhile, the newly generated image stream is processed by the TopK update arbitrator and stored in the partial image buffer, enabling depth-sparse memory management.

date arbitrator identifies the 30 most significant voxels along each scan line and arranges them in ascending depth order. A 54-bit bit mask is used to indicate the positions of valid voxels within the depth dimension.

The image buffer stores the real and imaginary values of these 30 selected voxels. In addition, the buffer maintains the corresponding bit mask and the index of the first valid voxel, enabling efficient reconstruction of the sparse voxel locations.

3.5.2 Synthetic Aperture Accumulation Schedule

Fig. 10 illustrates the scheduling strategy used for synthetic aperture accumulation. In the proposed system, five transmit events (TX0 – TX4) are employed to form a single synthetic aperture image.

Each transmit event generates a partial beamformed image corresponding to one transmit aperture. These partial images are produced sequentially by the beamformer and stored in

dedicated TX image buffers.

Instead of waiting for all transmit events to complete before starting the accumulation process, the proposed architecture adopts a streaming accumulation schedule. As shown in Fig. 10, the partial images from consecutive transmit events overlap in time. Once the required set of transmit images becomes available, the vector adder immediately performs coherent accumulation across the five transmit buffers to generate the synthetic aperture image.

After the first synthetic aperture image is generated from $TX_0 - TX_4$, the accumulation window slides forward by one transmit event. The next synthetic aperture image is therefore formed using $TX_1 - TX_0$ of the following cycle, followed by $TX_2 - TX_1$, and so on.

This sliding-window accumulation mechanism enables continuous streaming operation and maximizes hardware utilization. Since partial images are reused across multiple accumulation windows, the architecture avoids recomputing beamformed data and significantly reduces the memory bandwidth required for synthetic aperture processing.

This scheduling strategy allows each transmit image to be reused in multiple synthetic aperture reconstructions, improving computational efficiency while maintaining continuous streaming throughput.

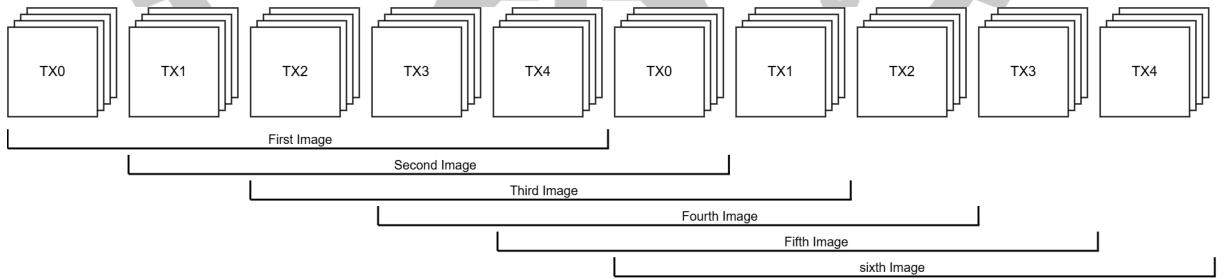


Figure 10: Scheduling of synthetic aperture accumulation. Each synthetic aperture image is formed by coherently accumulating five consecutive transmit images. The accumulation window slides forward for each new image, enabling continuous streaming operation.

3.5.3 Temporal Buffer

Fig. 12 illustrates the structure of the temporal buffer used to store the valid voxels of a scanline. The buffer is implemented as a shift-register array composed of 30 entries, where each entry stores a 24-bit voxel value corresponding to the complex beamformed signal.

Each register stage is preceded by a multiplexer controlled by the TopK update arbitrator. The arbitrator determines whether the incoming voxel value should be inserted into the buffer, discarded, or trigger a shift operation.

This shift-register architecture enables efficient hardware implementation since the data movement is realized through simple register transfers without requiring random memory access. Moreover, the pipeline-friendly structure allows the buffer to accept one voxel per cycle, matching the $II=1$ throughput of the upstream beamformer pipeline.

The temporal buffer stores only the most significant voxels within the active depth window. Combined with the valid-bit mask and the minimum depth coordinate, the full voxel locations can be reconstructed without explicitly storing the depth index of every voxel. This design significantly reduces the on-chip memory requirement and memory bandwidth compared to storing the entire 3-D voxel grid.

3.5.4 Streaming-Friendly Sparse Update

The sparse memory update mechanism is designed to operate in a fully streaming manner. Each incoming voxel is processed immediately by the TopK update arbitrator without stalling the upstream beamforming pipeline.

Since the temporal buffer is implemented using shift registers, the update operation only involves register shifts and simple comparisons. This design avoids random memory access and allows the sparse memory module to sustain the same $II=1$ throughput as the rest of the

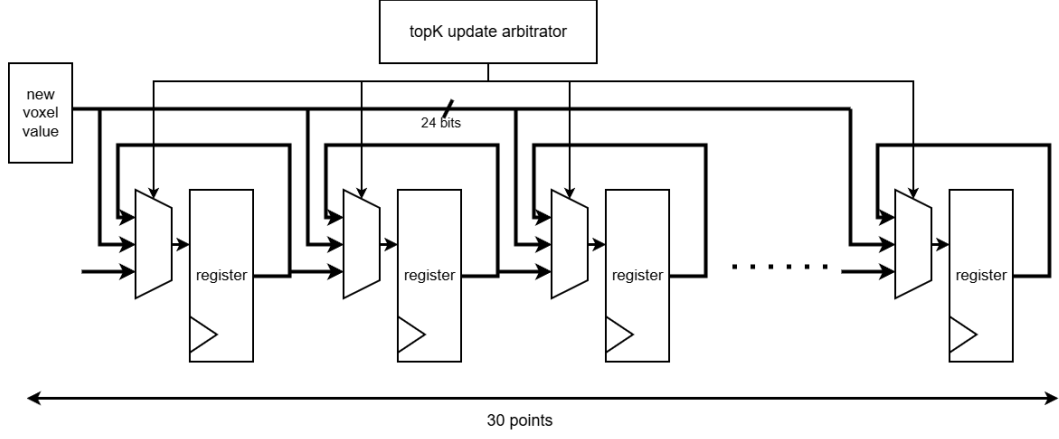


Figure 11: Structure of the temporal buffer used for sparse voxel storage. The buffer is implemented as a shift-register array containing 30 entries, where each entry stores a 24-bit voxel value. Controlled by the TopK update arbitrator, new voxel values can be inserted, shifted, or discarded, enabling efficient maintenance of the most significant voxels along each scanline.

processing pipeline.

3.5.5 Sparse Memory Update Policy

Fig. 12 illustrates the update policy of the proposed depth-sparse memory. Unlike a conventional Top- K sorter that ranks voxels by magnitude, the temporal buffer in our design stores valid voxels in *ascending depth order*. This ordering enables a compact location encoding: each scanline maintains a minimum depth index `min_coord` and a 54-bit valid bitmask, where the bitmask indicates which depth bins within the window $[\text{min_coord}, \text{min_coord} + 53]$ contain valid voxels. With this representation, the voxel coordinates can be recovered without storing an explicit depth index for every voxel, which would otherwise impose a prohibitive on-chip bandwidth requirement.

For each beamformed sample at the next depth, the system first applies a magnitude threshold test. Samples below the threshold are discarded. If the buffer is empty or not yet full, the voxel is inserted into the first available position while preserving the depth order, and the corresponding bit in the valid mask is set.

When the buffer is full, the incoming voxel is compared against the first stored voxel (i.e.,

the voxel at `min_coord`). This comparison is used to enforce a peak-containing window: as long as the incoming voxel magnitude remains higher than the leading voxel, the window is shifted forward by one depth bin (the buffer shifts left, the first voxel is discarded, and the new voxel is appended at the end). Meanwhile, `min_coord` is updated and the valid bitmask is shifted accordingly. Once the signal starts decaying and the newly arriving voxel magnitude becomes lower than that of the leading voxel, the shift operation stops, preventing the window from drifting beyond the peak region.

This policy maintains a compact depth window centered around the dominant echo response while preserving depth-ordered storage, thereby reducing both memory footprint and coordinate-storage bandwidth.

3.5.6 Sparse Memory Reconstruct Policy

To efficiently exploit depth sparsity in the proposed synthetic aperture imaging system, a selective reconstruction policy, as shown in Fig. 13, is adopted based on a finite state machine (FSM) and a validity mask. For each scanning coordinate, the system first determines whether the current depth falls within the predefined valid range. A validity bit mask is then checked to identify whether the corresponding depth location contains meaningful echo information.

If the valid bit is asserted, the system activates a read pointer to retrieve the stored beam-forming data and performs frame accumulation. Otherwise, no data access or accumulation is carried out, thereby avoiding unnecessary memory operations and computations. This selective processing mechanism ensures that only valid depth samples contribute to the final image formation, while invalid or empty regions are effectively skipped.

By eliminating redundant additions and memory accesses for invalid depth points, the proposed approach significantly reduces computational complexity and on-chip memory band-

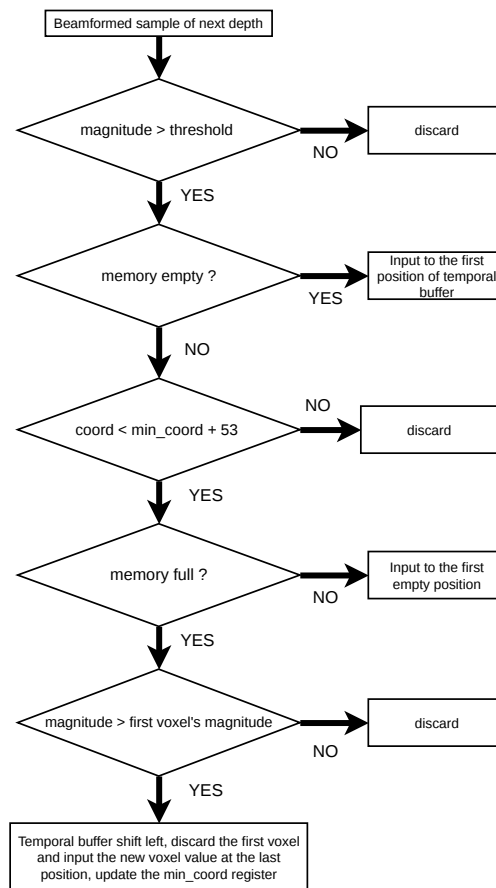


Figure 12: This figure demonstrate the algorithm of maintaining the top 30 points of a single scan line

width requirements. This design aligns with the inherent sparsity observed in ultrasonic imaging scenes, where only a small subset of depth samples along each ray contains useful signal information.

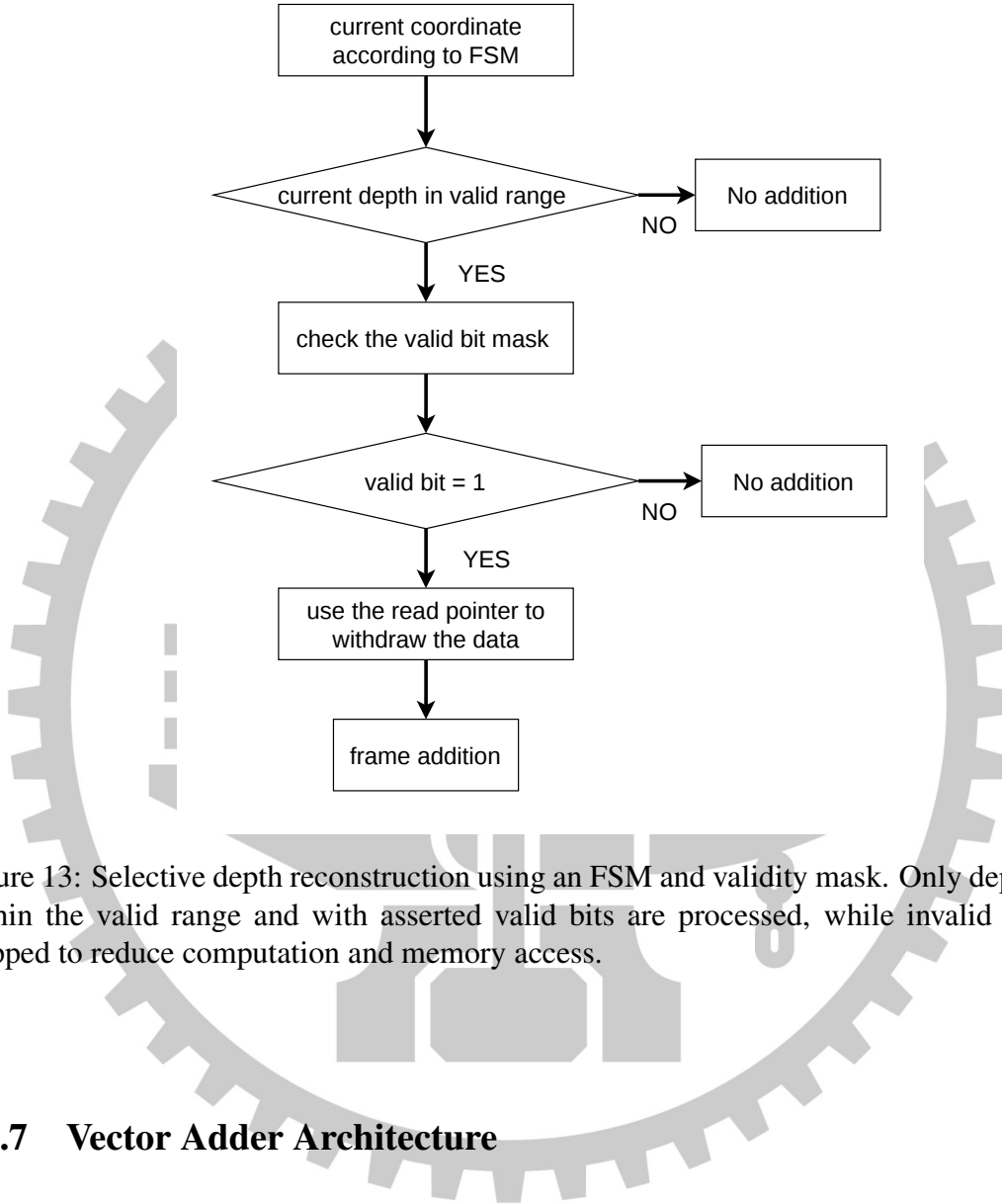


Figure 13: Selective depth reconstruction using an FSM and validity mask. Only depth samples within the valid range and with asserted valid bits are processed, while invalid regions are skipped to reduce computation and memory access.

3.5.7 Vector Adder Architecture

The vector adder performs coherent accumulation across the five transmit apertures. As shown in Fig. 9, the architecture first selects voxel values from the TX image buffers through a multiplexer network. The selected values are then accumulated using a pipelined adder tree.

The adder tree structure reduces the critical path delay compared with a serial accumulation scheme, enabling the accumulation process to match the $\Pi=1$ throughput of the beamformer

pipeline.



Chapter 4. Experiment Setup and Methodology

4.1 System Configuration

The configuration of the simulated ultrasonic imaging system is summarized in Table 2. The transducer array consists of 64 elements arranged in a two-dimensional sunflower sparse array. This array configuration is adopted to achieve a relatively large effective aperture while maintaining a limited number of transducer channels.

The center frequency of the ultrasonic transducer is set to 40 kHz, which is commonly used in airborne ultrasonic sensing systems. The acoustic propagation speed is assumed to be 340 m/s, corresponding to the speed of sound in air at room temperature.

The received RF signals are sampled with a sampling frequency of $3.125 \times 10^6/19$ Hz. This sampling rate is selected according to the system clock configuration used in the hardware implementation.

The imaging system is designed to cover a field of view (FOV) of $\pm 60^\circ$ in both the horizontal and vertical directions. The angular space is discretized into 69×69 scan lines. Along each scan line, the depth dimension is further divided into 284 sampling points. As a result, the reconstructed three-dimensional image consists of $69 \times 69 \times 284$ voxels.

The imaging depth ranges from 0.3 m to 2 m, which corresponds to the practical operating range of the airborne ultrasonic sensing system considered in this work.

Table 2: System configuration used in the simulation

Parameter	Value
Array type	2-D sunflower sparse array
Number of channels	64
Center frequency	40 kHz
Sampling frequency	$3.125 \times 10^6/19$ Hz
Speed of sound	340 m/s
Field of view	$\pm 60^\circ$
Number of scan lines	69×69
Depth samples	284
Depth range	0.3 m – 2 m
Total voxels	$69 \times 69 \times 284$

4.2 Point Target Simulation Using Field II

To evaluate the angular resolution of the proposed imaging system, the echo data are generated using the Field II ultrasound simulation toolbox. In the simulation environment, the target is modeled as an ideal point scatterer.

A point target represents a single isotropic reflector with a dimension much smaller than the acoustic wavelength. As a result, the reflected wave can be regarded as originating from a single spatial location. This type of target is commonly used in ultrasound imaging research to characterize the point spread function (PSF) and the spatial resolution of the imaging system.

In the Field II simulation, the transmitted ultrasonic wave propagates from the transducer array toward the point scatterer. The scattered wave is then received by all array elements and converted into RF signals according to the acoustic propagation model implemented in Field II. Since the scatterer is ideal and does not introduce any extended structure, the resulting image response directly reflects the intrinsic beam pattern of the imaging system.

Therefore, the reconstructed image of a point target effectively represents the system's point spread function. The width of the mainlobe in the angular response corresponds to the angular resolution of the imaging system, while the sidelobe levels indicate the amount of off-axis

interference.

Since the simulated target is an ideal point scatterer, the reconstructed response directly corresponds to the system point spread function.

4.3 Algorithm Development Flow

Fig. 14 illustrates the development flow of the proposed system. The design process begins with acoustic data generation using the Field II simulator, which provides realistic channel responses for the target scenario. These simulated datasets are then processed using a MATLAB-based synthetic aperture (SA) beamforming model to evaluate the imaging performance.

It should be noted that the synthetic aperture imaging algorithm itself is not a contribution of this work. The SA beamforming principle has been widely studied and established in prior literature. In this thesis, the SA algorithm is adopted as the baseline imaging method.

The primary objective of this stage is to establish a software reference model that can be used for validating the correctness of the hardware implementation. The MATLAB simulation evaluates whether the desired imaging resolution is achieved. If the resolution specification is not satisfied, the algorithm parameters or system configuration are adjusted and the simulation is repeated.

Once the imaging performance satisfies the resolution requirements, the algorithm is translated into a hardware-oriented implementation using high-level synthesis (HLS). During this step, several modifications are introduced to make the algorithm more suitable for hardware realization. These modifications aim to reduce computational complexity, improve dataflow efficiency, and enable pipelined execution in the FPGA architecture.

The resulting HLS model is first verified through C simulation (CSIM), and the generated output is compared with the MATLAB reference through visualization. If the resolution speci-

cation is not preserved after the hardware-oriented modifications, the hardware model is refined and the verification loop is repeated. Once the algorithm behavior matches the reference model and satisfies the resolution requirement, the algorithm development stage is completed and the design proceeds to hardware synthesis and implementation.

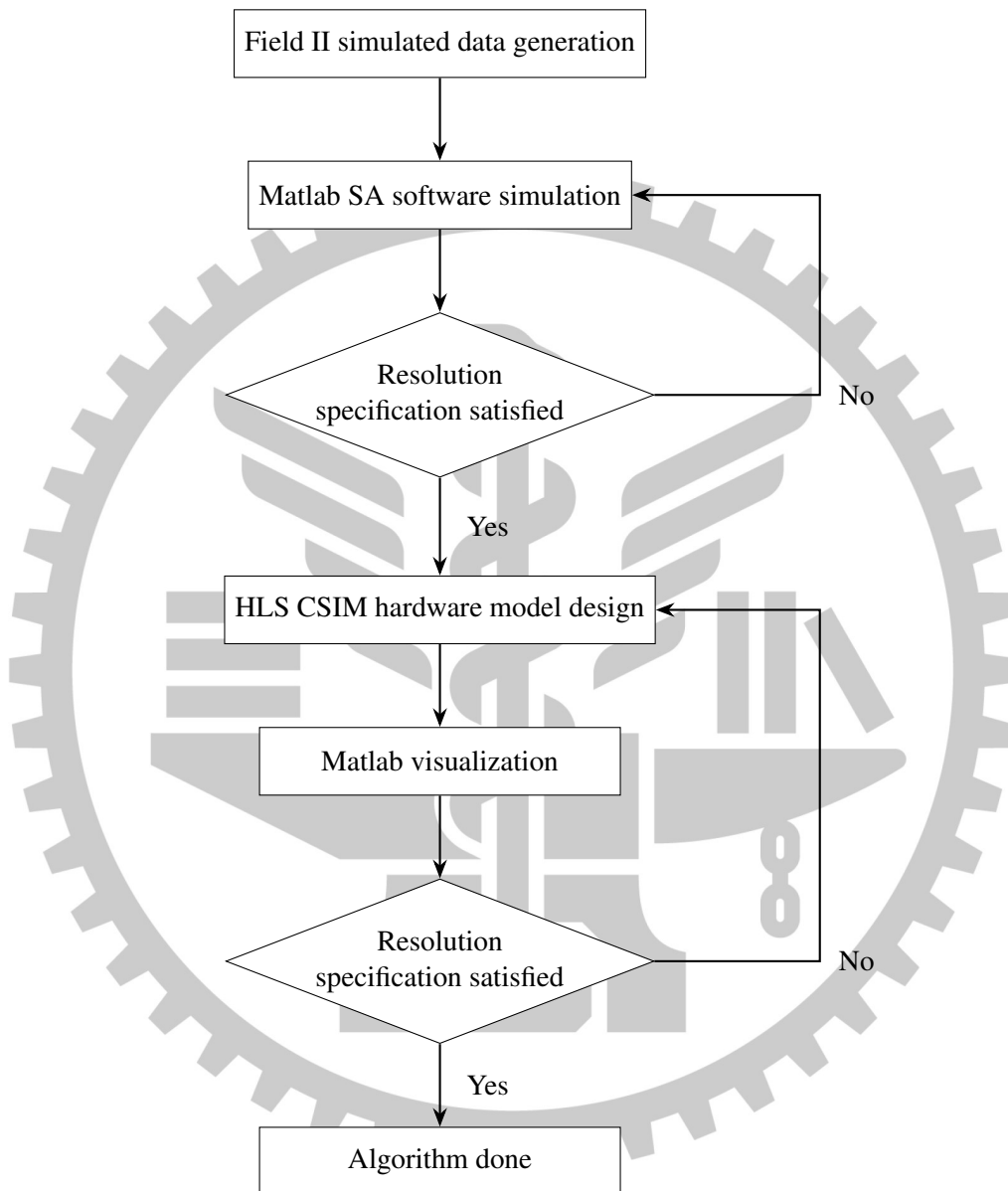


Figure 14: Algorithm development flow for the proposed synthetic aperture imaging system.

4.4 Hardware Implementation Flow

Fig. 15 illustrates the hardware implementation procedure of the proposed system. After the algorithm model is validated in the software stage, the design is translated into a hardware-oriented model using high-level synthesis (HLS).

The hardware development begins with the HLS C simulation (CSIM) model design, where the algorithm is rewritten in a hardware-friendly manner. This step focuses on introducing pipeline structures, dataflow scheduling, and parallel computation to enable efficient FPGA implementation.

Once the hardware model is established, the design proceeds to C synthesis (CSYN). During this stage, the HLS tool generates the corresponding RTL and provides resource utilization and timing estimation. The obtained results are then evaluated against the target performance specifications. If the design does not satisfy the required performance constraints, such as latency or resource usage, the hardware model is refined and the synthesis process is repeated.

When the synthesized design satisfies the performance requirements, co-simulation (COSIM) is performed to verify the functional equivalence between the generated RTL and the original C model. After successful verification, the design proceeds to the final RTL synthesis (VSYN) stage.

Finally, the synthesized hardware is again evaluated with respect to the target performance specifications. If the requirements are met, the hardware implementation process is completed. Otherwise, the design is iteratively refined until the desired performance is achieved.

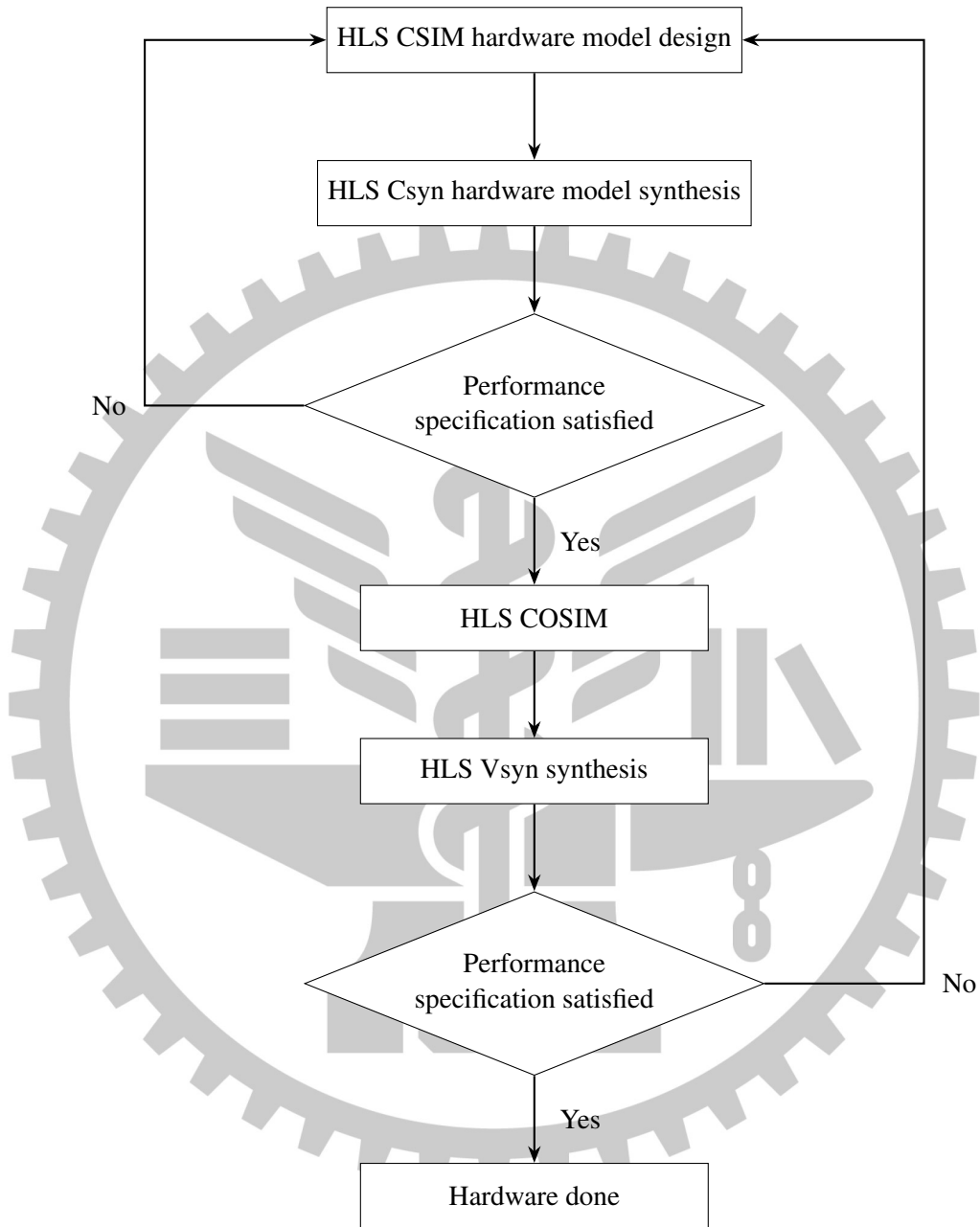


Figure 15: Hardware implementation and verification flow of the proposed beamforming architecture.

4.5 Performance Evaluation Metrics

The imaging performance of the proposed system is evaluated using the angular response obtained from point target simulations. Since the target is modeled as an ideal point scatterer, the reconstructed image directly corresponds to the point spread function (PSF) of the imaging system. Therefore, the characteristics of the PSF can be used to quantitatively evaluate the performance of the system.

4.5.1 Angular Resolution

The angular resolution of the imaging system is evaluated using the -6 dB beamwidth of the mainlobe. The beam pattern is first normalized with respect to the maximum amplitude of the response. The angular resolution is then defined as the angular distance between the two points where the normalized magnitude drops to -6 dB from the peak value.

Mathematically, if $B(\theta)$ denotes the normalized beam response, the angular resolution $\Delta\theta$ is defined as

$$B(\theta) = -6 \text{ dB} \quad (4.1)$$

and

$$\Delta\theta = \theta_2 - \theta_1 \quad (4.2)$$

where θ_1 and θ_2 represent the two angles at which the response crosses the -6 dB level. A smaller beamwidth indicates a higher angular resolution.

4.5.2 Sidelobe Level

In addition to the mainlobe width, the sidelobe level is also used to evaluate the imaging quality. The sidelobe level represents the relative magnitude of off-axis responses compared to the mainlobe peak. High sidelobe levels may introduce false targets or image artifacts.

The sidelobe level is measured as the maximum amplitude of the sidelobes relative to the peak value of the mainlobe in the normalized beam pattern.

4.5.3 Comparison Between Conventional and Synthetic Aperture Imaging

To evaluate the benefit of the synthetic aperture (SA) technique, the beam patterns generated by individual transmit events are compared with the final beam pattern obtained after synthetic aperture accumulation. The improvement in angular resolution is quantified by comparing the -6 dB beamwidth before and after synthetic aperture summation.

Chapter 5. Performance Evaluation

5.1 Resolution Analysis

To evaluate the angular resolution improvement introduced by the synthetic aperture (SA) approach, the beampatterns of individual transmit events and the synthetic aperture summation are compared. Fig. 20, Fig. 23 and Fig. 16, Fig. 23 show the normalized angular responses at different depths for two set of target locations.

For each transmit event (Tx01–Tx05), the beamformer produces a low-resolution response since only a single transmit aperture is used. The corresponding mainlobe widths are relatively large, typically around 4° – 7° depending on the depth and steering direction.

After applying synthetic aperture accumulation, the responses from multiple transmit events are coherently summed. The resulting beampattern, denoted as *Sum* in the figures, exhibits a significantly narrower mainlobe. The measured -6 dB beamwidth is approximately 2.5° – 3.3° , which is substantially smaller than that of the individual transmit responses.

This reduction in beamwidth is achieved because synthetic aperture imaging effectively synthesizes a larger transmit aperture by combining multiple transmit events. As a result, the spatial resolution in the angular domain is improved while maintaining the same physical array size.

5.1.1 Central Target without Coherent Factor

To clearly highlight the contribution of synthetic aperture (SA) beamforming, the results without applying the coherence factor (CF) are presented first. Unlike a conventional baseline comparison, these results are intended to isolate and reveal the intrinsic performance of the beamforming schemes, particularly the effect of SA on angular resolution and sidelobe characteristics.

Tables 3 and 4 summarize the measured mainlobe width and peak sidelobe level at different depths for two target locations. From these results, it can be observed that SA consistently achieves a narrower mainlobe compared to individual transmit (TX) cases, demonstrating its effectiveness in improving spatial resolution. However, the sidelobe levels remain relatively high, as no additional coherence-based weighting is applied.

It is important to note that the omission of CF in this stage is intentional, as CF is not the primary focus of this work. Instead, it is introduced later as a complementary technique to further enhance image quality, particularly in terms of sidelobe suppression and contrast improvement.

Table 3: Measured mainlobe width (in degrees) and peak sidelobe level (in dB) without coherence factor (w/o CF) at different depths for a point target located at $(0^\circ, 0^\circ)$.

Depth	TX0 (°/dB)	TX1 (°/dB)	TX2 (°/dB)	TX3 (°/dB)	TX4 (°/dB)	SA (°/dB)
0.5 m	7.11 / -10.4	6.85 / -10.1	6.68 / -9.5	7.20 / -10.2	7.20 / -10.3	2.93 / -10.4
1.0 m	6.41 / -11.1	6.08 / -12.9	5.99 / -11.7	4.70 / -10.6	4.67 / -9.4	3.06 / -14.9
1.5 m	4.74 / -10.6	6.20 / -11.6	6.16 / -11.3	4.96 / -9.9	4.99 / -9.5	2.86 / -15.5

5.1.2 off-axis Target without Coherent Factor

In addition to the on-axis case, the performance of the proposed beamforming scheme is further evaluated at an off-axis target location. The corresponding results without applying the

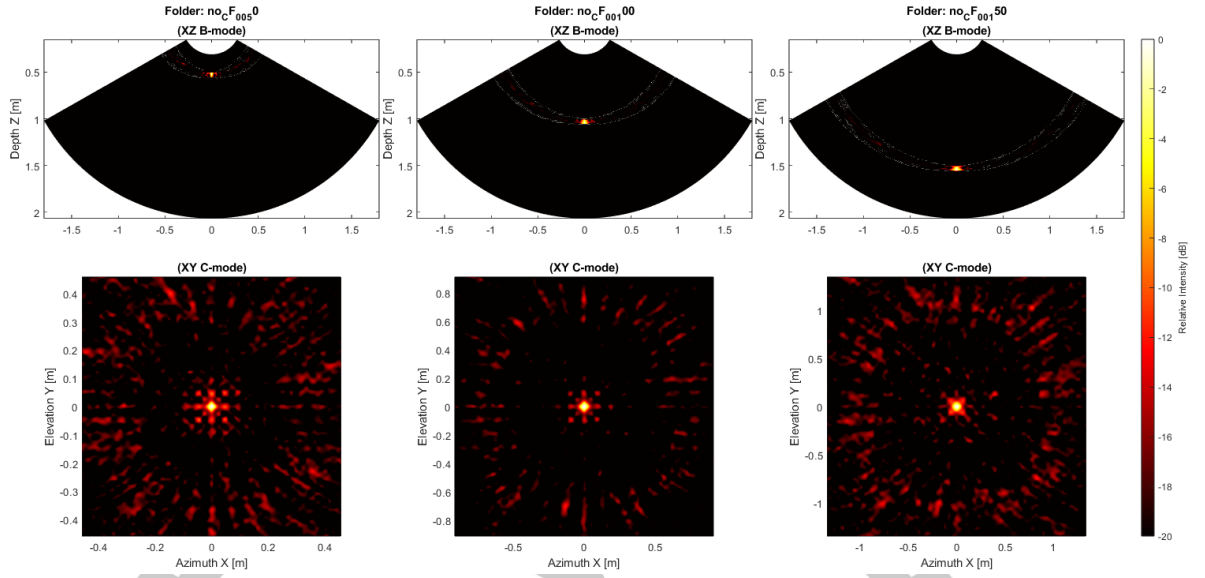


Figure 16: Beamformed images without coherence factor (w/o CF) for an on-axis point target located at $(0^\circ, 0^\circ)$. The top row shows the XZ B-mode images, and the bottom row shows the corresponding XY C-mode slices. From left to right, the target depth increases from 0.5 m to 1.5 m. The results illustrate the intrinsic performance of synthetic aperture (SA) beamforming, where a focused mainlobe is maintained across different depths. However, noticeable sidelobes and background clutter are present due to the absence of coherence-based weighting.

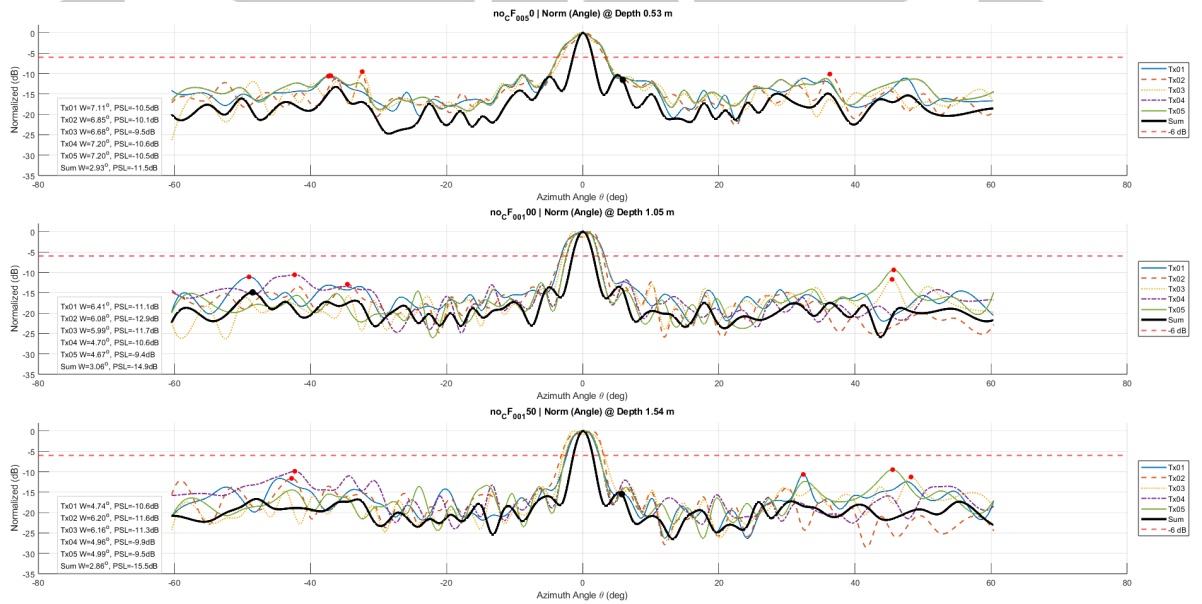


Figure 17: Normalized beampatterns without coherence factor (w/o CF) for an on-axis target at $(0^\circ, 0^\circ)$ at depths from 0.5 m to 1.5 m (top to bottom).

coherence factor (CF) are summarized in Table 4.

Compared to the on-axis scenario, beamforming performance generally degrades under off-axis conditions due to increased phase mismatch and reduced aperture coherence. This effect is particularly evident in the single-transmit (TX) cases, where both mainlobe broadening and elevated sidelobe levels can be observed.

Nevertheless, synthetic aperture (SA) beamforming still maintains a clear advantage in this scenario. The results show that SA consistently achieves a narrower mainlobe and improved angular resolution compared to individual TX cases, even under off-axis conditions. This demonstrates the robustness of SA in preserving spatial resolution across different target positions.

It is worth noting that, similar to the on-axis analysis, no coherence factor is applied at this stage in order to isolate the intrinsic behavior of SA. The relatively higher sidelobe levels observed here further motivate the use of CF, which will be introduced in the subsequent section to enhance sidelobe suppression and overall image contrast.

Table 4: Measured mainlobe width (in degrees) and peak sidelobe level (in dB) without coherence factor (w/o CF) at different depths for an off-axis point target.

Depth	TX0 (°/dB)	TX1 (°/dB)	TX2 (°/dB)	TX3 (°/dB)	TX4 (°/dB)	SA (°/dB)
0.5 m	8.60 / -9.4	6.74 / -9.0	9.05 / -8.9	8.92 / -9.6	8.02 / -7.8	3.82 / -14.4
1.0 m	8.80 / -10.7	10.10 / -8.6	7.75 / -10.6	8.80 / -8.2	8.99 / -10.1	3.11 / -14.7
1.5 m	8.34 / -8.9	9.78 / -10.1	8.31 / -10.8	8.99 / -10.0	9.72 / -8.4	3.77 / -14.0

5.1.3 Central Target with Coherent Factor

Table 5: Measured mainlobe width (in degrees) and peak sidelobe level (in dB) at different depths for a point target located at $(0^\circ, 0^\circ)$.

Depth	TX0 (°/dB)	TX1 (°/dB)	TX2 (°/dB)	TX3 (°/dB)	TX4 (°/dB)	SA (°/dB)
0.5 m	3.54 / -29.1	2.99 / -30.6	3.03 / -29.5	3.59 / -28.3	3.61 / -29.2	2.30 / -31.7
1.0 m	4.26 / -33.0	3.45 / -29.4	3.64 / -35.6	4.24 / -32.7	4.24 / -32.7	2.56 / -37.2
1.5 m	3.71 / -28.8	4.88 / -29.5	4.69 / -34.0	4.72 / -36.2	4.72 / -36.2	2.49 / -41.2

Table 5 summarizes the measured -6 dB mainlobe width for a point target located at $(0^\circ, 0^\circ)$

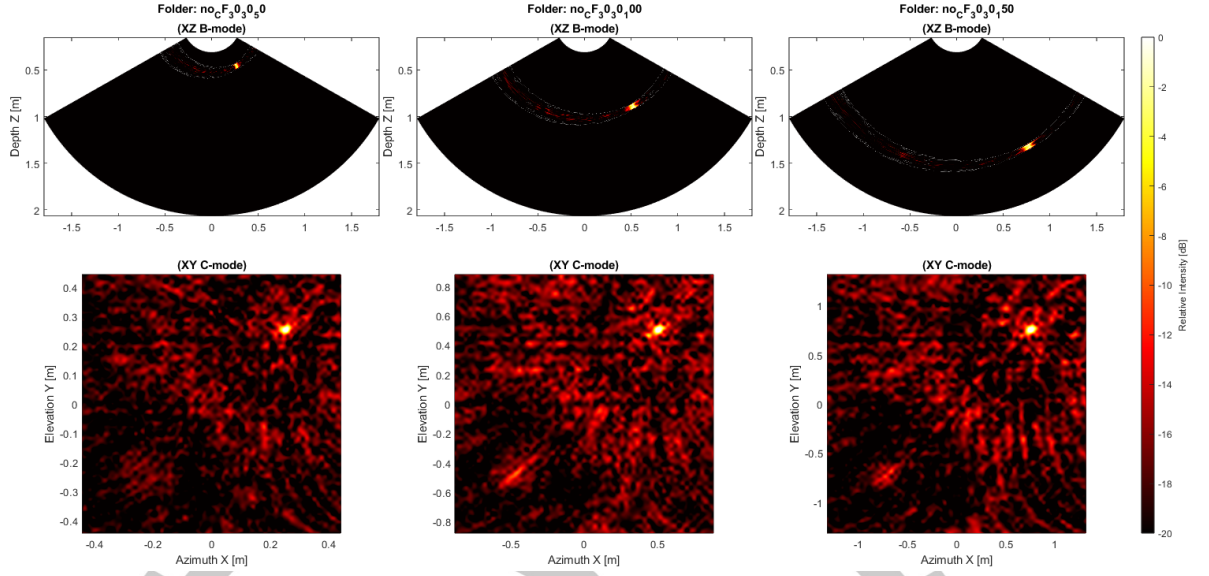


Figure 18: Beamformed images without coherence factor (w/o CF) for an off-axis point target. The top row shows the XZ B-mode images, and the bottom row shows the corresponding XY C-mode slices. From left to right, the target depth increases from 0.5 m to 1.5 m. Compared to the on-axis case, increased sidelobes and clutter can be observed, especially in the C-mode images, due to reduced aperture coherence under off-axis conditions. Nevertheless, synthetic aperture (SA) beamforming preserves a focused mainlobe, demonstrating its robustness across different target positions.

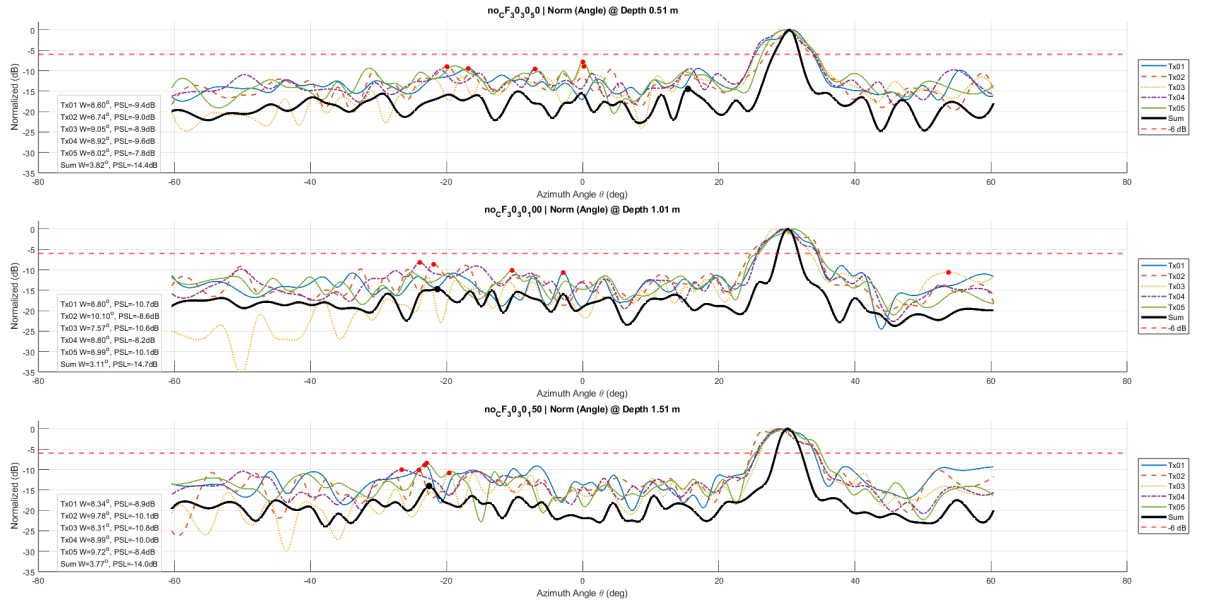


Figure 19: Normalized beampatterns without coherence factor (w/o CF) for an off-axis target at depths from 0.5 m to 1.5 m (top to bottom).

under different depths. The table compares the angular resolution obtained from individual transmit events (TX0–TX4), corresponding to single-transmit imaging, and the final synthetic aperture (SA) reconstruction.

For single-transmit imaging, the effective aperture is limited to a single transmission event, resulting in relatively wider mainlobes and reduced angular resolution. As shown in the table, the measured -6 dB beamwidths of TX0–TX4 range from approximately 3° to 5° depending on the depth.

After combining multiple transmit events using synthetic aperture accumulation, the effective aperture is significantly increased. Consequently, the SA reconstruction consistently achieves a narrower mainlobe, with beamwidths around 2.3° – 3.0° across the evaluated depths.

In addition to the improvement in mainlobe width, the proposed synthetic aperture reconstruction also reduces sidelobe artifacts. In the single-transmit imaging results, as shown in Fig. 22, noticeable sidelobe patterns and background artifacts appear in the C-mode images due to the limited effective aperture and incomplete spatial sampling.

After combining multiple transmit events through synthetic aperture accumulation, the sidelobe level is significantly suppressed and the background artifacts are reduced as shown in Fig. 21. As a result, the target response becomes more concentrated and the overall image contrast is improved.

This result demonstrates that synthetic aperture beamforming effectively improves angular resolution compared with single-transmit imaging.

5.1.4 Off-axis Target with Coherent Factor

Table 6 summarizes the measured -6 dB mainlobe width for a point target located at $(30^\circ, 30^\circ)$. The table compares the beamwidth obtained from individual transmit events (TX0–

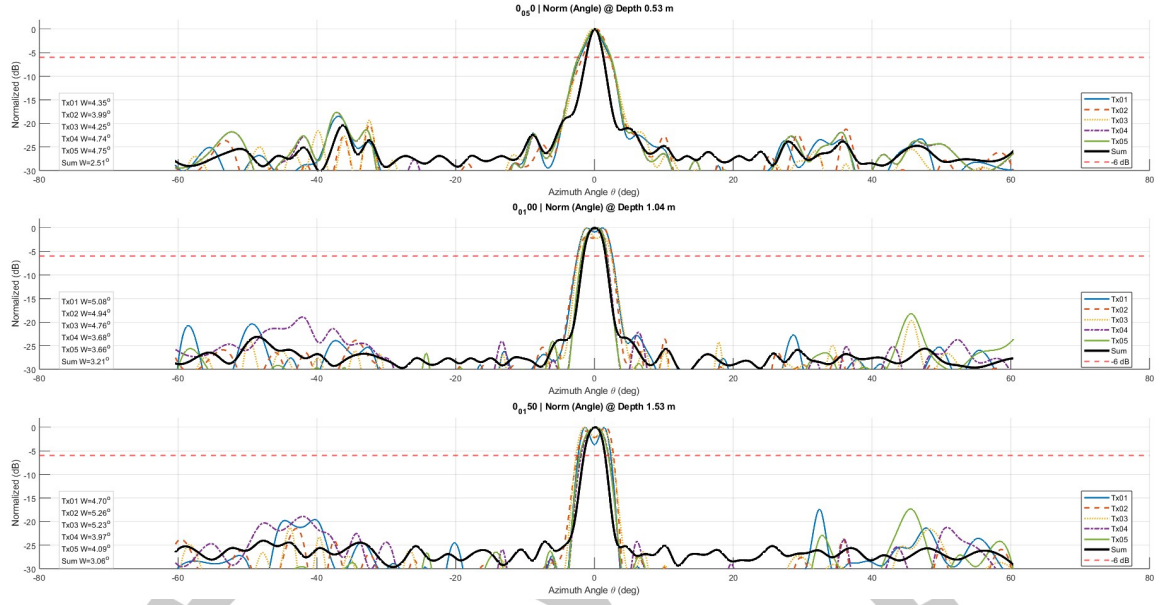


Figure 20: Angular response comparison between individual transmit events and synthetic aperture summation for an on-axis target. The synthetic aperture response (Sum) achieves a significantly narrower mainlobe compared with the responses of individual transmit events.

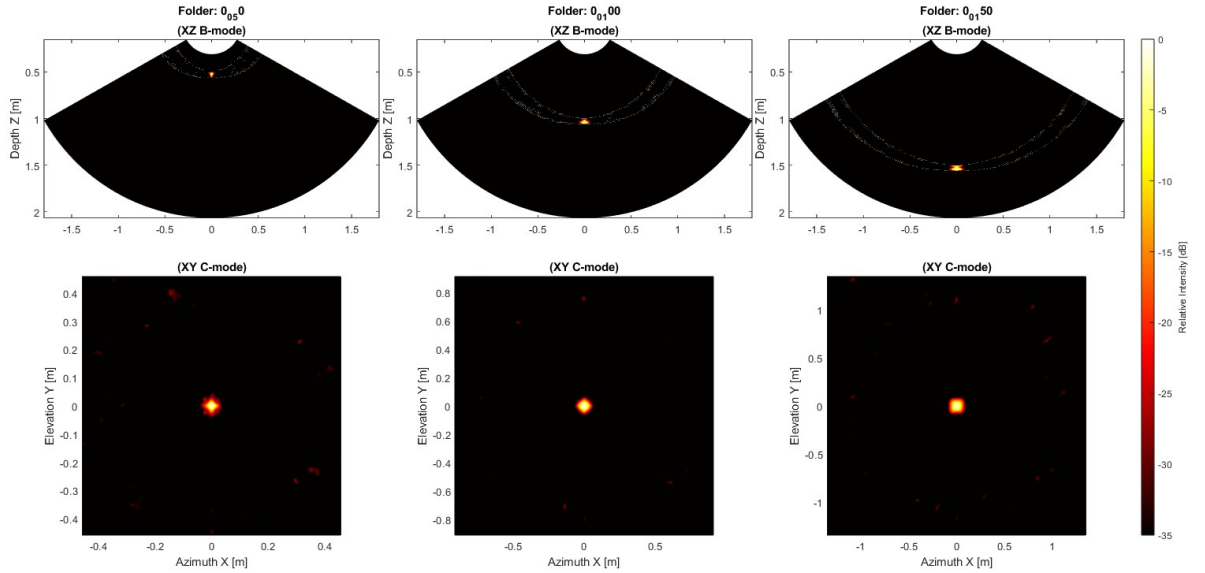


Figure 21: Reconstructed images of a simulated point target located on the central axis of the array ($0^\circ, 0^\circ$). The upper row presents XZ B-mode slices and the lower row shows the corresponding XY C-mode images. The three columns correspond to depths of 0.5 m, 1.0 m, and 1.5 m. The reconstructed responses are centered along the array axis, indicating accurate depth localization and consistent spatial focusing of the proposed imaging system across different ranges.

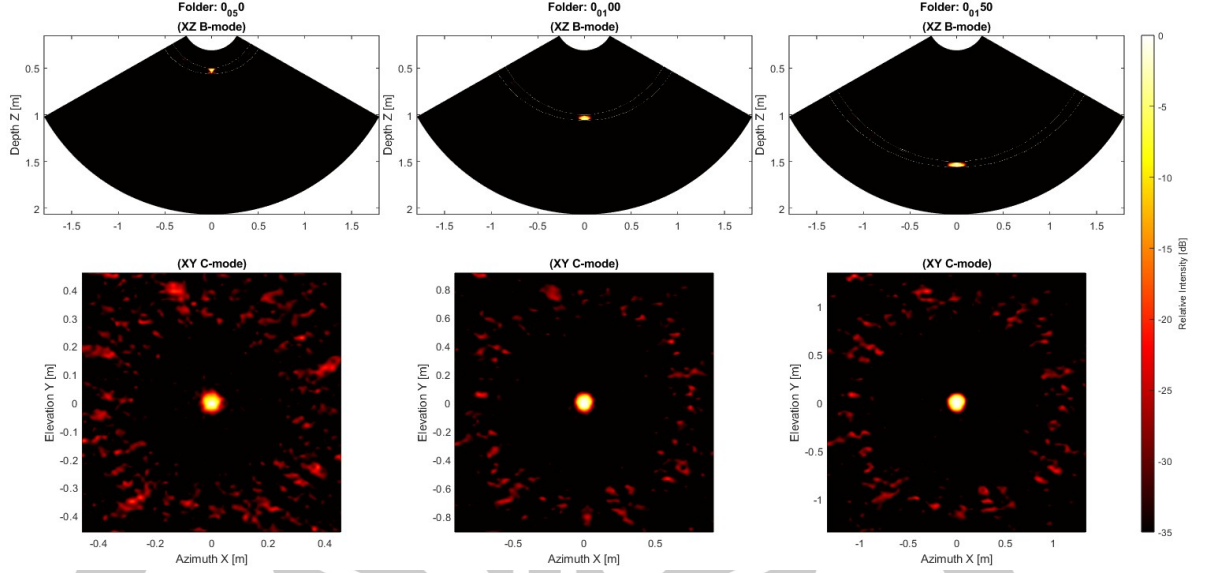


Figure 22: Single-transmit (single-TX) reconstruction results for a point target located on the central axis of the array ($0^\circ, 0^\circ$). The upper row shows XZ B-mode slices and the lower row shows the corresponding XY C-mode images. The three columns correspond to target depths of 0.5 m, 1.0 m, and 1.5 m. The reconstructed echo appears near the center of the image; however, noticeable background artifacts and weaker spatial focusing can be observed due to the limited aperture of single-transmit imaging.

Table 6: Measured mainlobe width (in degrees) and peak sidelobe level (in dB) at different depths for a point target located at ($30^\circ, 30^\circ$).

Depth	TX0 ($^\circ$ /dB)	TX1 ($^\circ$ /dB)	TX2 ($^\circ$ /dB)	TX3 ($^\circ$ /dB)	TX4 ($^\circ$ /dB)	SA ($^\circ$ /dB)
0.5 m	4.23 / -25.3	3.72 / -23.4	3.34 / -25.9	3.18 / -23.1	5.33 / -21.5	2.68 / -32.2
1.0 m	5.00 / -26.1	5.22 / -31.5	4.83 / -27.9	3.92 / -18.5	5.75 / -26.2	2.90 / -32.4
1.5 m	5.29 / -25.1	5.14 / -22.8	5.08 / -22.3	4.37 / -24.1	4.60 / -22.1	3.00 / -31.7

TX4), corresponding to single-transmit imaging, and the final synthetic aperture (SA) reconstruction.

Compared with the on-axis case, the beamwidth of individual transmit events becomes slightly wider due to the off-axis steering and the reduced effective aperture in the steered direction. As a result, single-transmit imaging exhibits degraded angular resolution for off-axis targets.

Nevertheless, the synthetic aperture (SA) reconstruction still produces a significantly narrower mainlobe across all evaluated depths. By coherently combining multiple transmit events, the effective aperture is enlarged, which improves spatial focusing capability.

This result indicates that the proposed SA imaging approach effectively enhances angular resolution not only for on-axis targets but also for off-axis targets within the imaging field of view.

In addition to the improvement in angular resolution, the synthetic aperture reconstruction also suppresses sidelobe artifacts. In the single-transmit imaging results, strong sidelobe patterns and background clutter can be observed in the C-mode images, as shown in Fig. 25 due to the limited effective aperture of a single transmission.

After combining multiple transmit events through synthetic aperture accumulation, the sidelobe level is significantly reduced and the background artifacts are suppressed. As a result, the target response appears more concentrated and the overall image contrast is improved, as shown in Fig. 24.

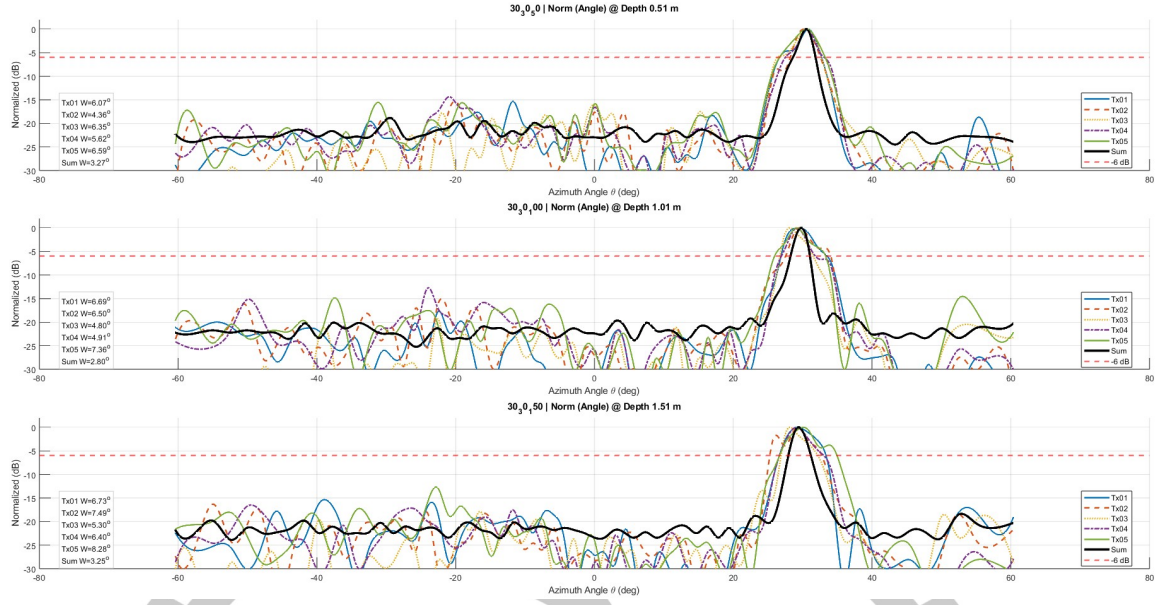


Figure 23: Angular response comparison for an off-axis target. Although the individual transmit responses exhibit wider beamwidths and higher sidelobes, the synthetic aperture summation effectively reduces the mainlobe width and improves angular resolution.

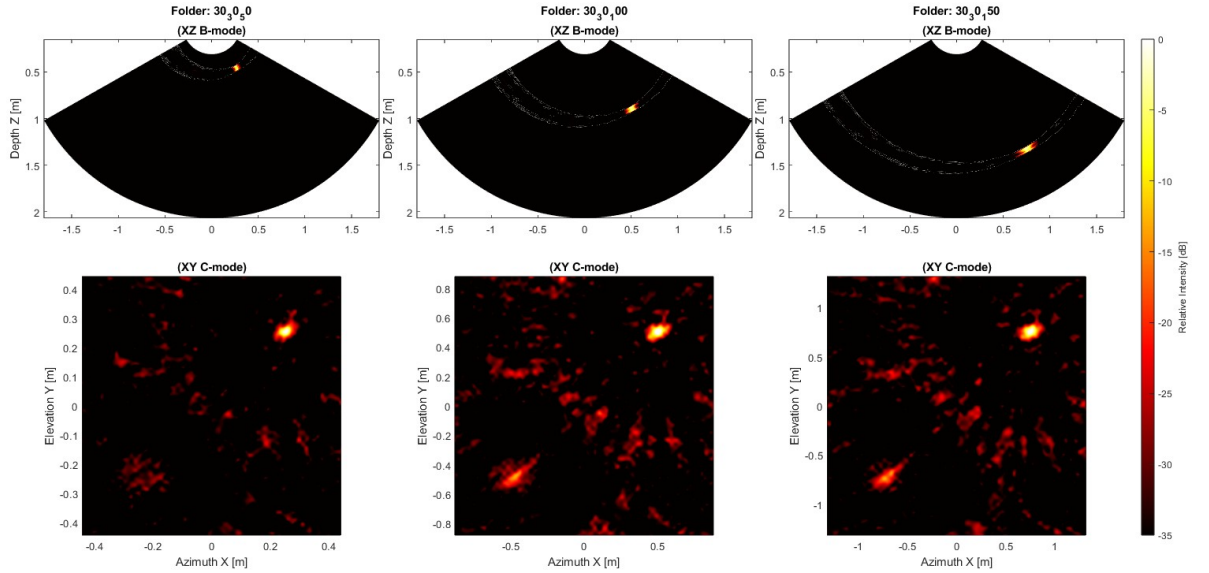


Figure 24: Reconstructed images of a simulated point target located at approximately $(30^\circ, 30^\circ)$ in the field of view. The upper row shows XZ B-mode slices, while the lower row shows corresponding XY C-mode images. From left to right, the target depths are 0.5 m, 1.0 m, and 1.5 m. The reconstructed echoes appear as localized high-intensity responses at the expected spatial locations, demonstrating that the proposed synthetic aperture beamforming maintains accurate target localization and spatial focusing performance for off-axis targets.

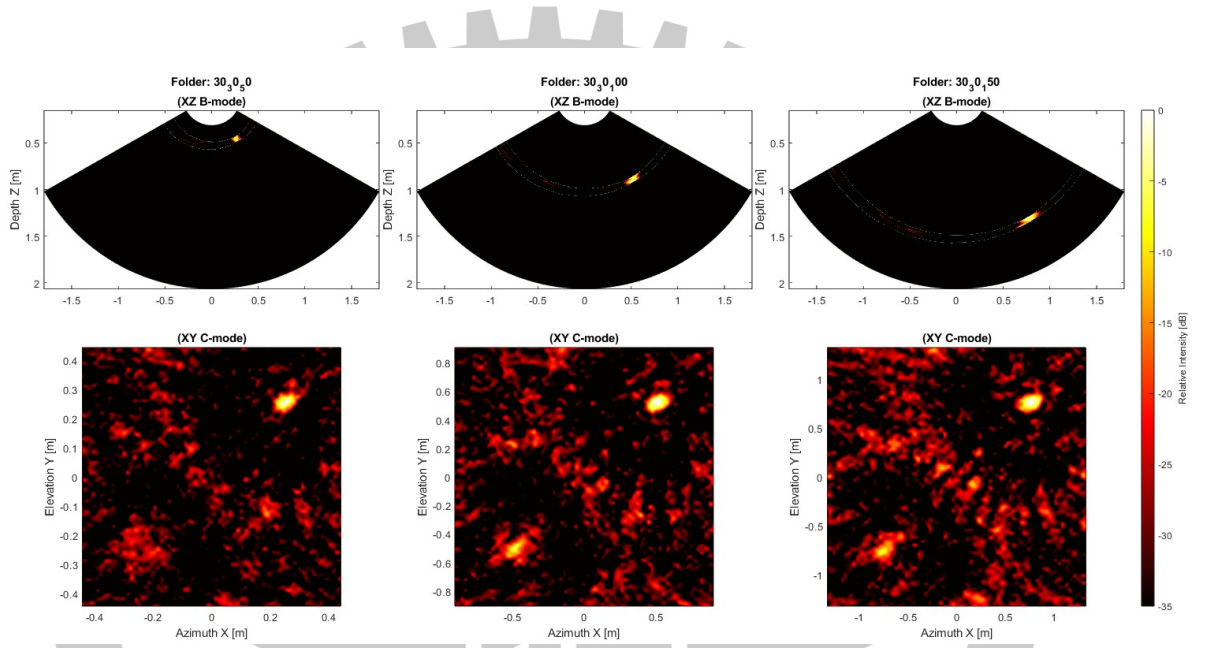


Figure 25: Single-transmit (single-TX) reconstruction results for a point target located at approximately $(30^\circ, 30^\circ)$ in the imaging volume. The upper row shows XZ B-mode slices, while the lower row presents the corresponding XY C-mode images. From left to right, the target depths are 0.5 m, 1.0 m, and 1.5 m. The target is visible at the expected spatial location, but the reconstructed images exhibit stronger background artifacts and reduced focusing performance compared with the synthetic aperture reconstruction.

5.2 Performance Analysis

5.2.1 Memory Optimization and Latency Analysis

One of the major challenges in implementing synthetic aperture (SA) beamforming on FPGA platforms is the large memory requirement. In synthetic aperture imaging, multiple transmit events must be combined to form the final image. Consequently, the partial image generated from each transmit event must be stored before the final summation is performed.

In the conventional approach, each transmit event produces a complete 3-D image frame, which must be temporarily stored in memory. Since every frame corresponds to a full volumetric image, the required memory capacity becomes extremely large. Furthermore, when the intermediate frames are stored in external DRAM, the accumulation process requires frequent memory accesses through the AXI interface. This introduces additional latency overhead due to the limited burst capability and access latency of DRAM.

To address this issue, the proposed architecture adopts an indexed sparse memory strategy. Instead of storing the entire volumetric image for each transmit event, only the active pixels participating in the synthetic aperture accumulation are recorded together with their corresponding coordinates. As illustrated in Fig. 9, the beamforming pipeline generates voxel values through the delay calculation and beamforming stages. The resulting voxel values are then accumulated using a vector adder while the sparse memory stores only the valid voxel entries.

This design significantly reduces the memory footprint and enables the entire accumulation process to remain on-chip. By avoiding the use of external DRAM, the system eliminates the latency overhead associated with DRAM access and allows the streaming pipeline to maintain continuous data processing.

To evaluate the effectiveness of the proposed sparse memory architecture, the memory usage

and latency are compared with the baseline design that relies on conventional frame storage.

The results are summarized in Table 7.

Table 7: Memory usage and latency comparison between the original architecture and the optimized sparse memory architecture

Design	BRAM	Min Latency	Avg Latency	Max Latency
Original	148.52%	1918216	1918221	1918241
Optimized	83.77%	1823005	1823010	1823030

The results show that the proposed sparse memory architecture significantly reduces the on-chip memory usage. The BRAM consumption is reduced from 148.52% to 83.77%, making the design feasible for the target FPGA platform.

In addition to memory reduction, the optimized architecture also decreases the processing latency. This improvement is mainly attributed to the elimination of external DRAM accesses. Since AXI streaming from DRAM cannot always sustain continuous burst transfers, memory stalls may occur during the accumulation process. By keeping the sparse memory on-chip, the proposed architecture avoids these stalls and allows the pipeline to operate more efficiently.

These results demonstrate that the sparse memory strategy effectively reduces both memory usage and system latency, making the proposed architecture more suitable for real-time synthetic aperture imaging applications.

5.2.2 Computation Optimization and Throughput Improvement

In addition to memory consumption, computational complexity is another major challenge in implementing synthetic aperture (SA) beamforming on FPGA platforms. The overall processing pipeline consists of three major stages, including delay calculation, beamforming, and synthetic aperture accumulation. Among these stages, delay calculation is particularly computation-intensive because it requires distance evaluation for every scan point and every receive channel.

To address this issue, two optimization strategies are adopted in the proposed architecture.

First, a linear approximation method is introduced to simplify the delay calculation. Second, a streaming dataflow architecture is implemented to connect the delay calculation, beamforming, and vector accumulation stages, thereby improving the overall pipeline throughput.

5.2.3 Resource Reduction by Linear Delay Approximation

The delay calculation originally requires evaluating the Euclidean distance between the imaging point and the receiver position, which involves several multiplication and square-root operations. Such computations are expensive for FPGA implementation.

To reduce the hardware complexity, the proposed architecture adopts a linear approximation for the receive delay calculation. This approximation significantly simplifies the required arithmetic operations and allows the delay computation to be implemented using lightweight arithmetic logic.

Table 8 compares the hardware resource utilization between the original delay computation and the proposed linear approximation method.

Table 8: Hardware resource utilization comparison for delay calculation

Design	FF	DSP	BRAM	LUT
Original	76860 (14.02%)	329 (13.06%)	3 (0.16%)	240793 (87.86%)
Approximated	5549 (1.02%)	388 (15.40%)	64 (3.51%)	11313 (4.13%)

As shown in Table 8, the proposed approximation method significantly reduces the usage of look up table(LUT) resources. In particular, the LUT utilization decreases dramatically from 240793 to 11313, which represents a substantial reduction in combinational logic complexity. Although the DSP and BRAM usage increase slightly due to the introduction of lookup tables, the overall resource utilization becomes much more suitable for FPGA implementation.

5.2.4 Throughput Improvement Using Dataflow Architecture

In the original sequential implementation, the delay calculation, beamforming, and synthetic aperture accumulation stages are executed sequentially. This results in a long processing latency since each stage must complete before the next stage begins.

To improve the processing throughput, the proposed architecture connects the three stages using a streaming dataflow structure. By inserting dataflow FIFOs between stages, the delay calculation, beamforming, and vector addition modules can operate concurrently in a pipelined manner. The execution timeline is shown in Fig. 4.

In addition to the coarse-grain dataflow pipeline, a fine-grain pipeline is also employed, as shown in Fig. 5 inside the processing flow. In this scheme, different voxels are processed in an overlapped manner. This fine-grain pipelining further improves the throughput of the system and enables continuous streaming operation of the beamforming pipeline, allowing the system to achieve the initial interval of 1 cycle.

Table 9 shows the latency comparison between the sequential implementation and the proposed dataflow architecture.

Table 9: Latency comparison between sequential and dataflow architectures

Design	Latency (cycles)	FPS
Original	4,555,769	27.44
Dataflow	1,823,010	68.56

The results show that the proposed dataflow architecture reduces the overall processing latency from 4,555,769 cycles to 1,823,010 cycles, achieving approximately $2.46\times$ performance improvement. This improvement is mainly attributed to the concurrent execution of multiple pipeline stages, which significantly increases the hardware utilization efficiency.

5.2.5 Discussion and Comparison with Prior Work

Based on Table 9 and the beampattern analysis, the proposed dataflow architecture significantly reduces latency and improves frame rate, demonstrating the effectiveness of the streaming and pipelined design.

In terms of imaging performance, the synthetic aperture (SA) approach achieves improved spatial resolution and sidelobe suppression, while the incorporation of coherent factor (CF) further enhances contrast, which is critical in low-SNR air-coupled environments.

Compared with [23], which employs fully parallel per-voxel beamforming for real-time performance, the proposed method achieves better throughput by adopting hybrid delay calculator. Additionally, in contrast to GPU-assisted systems such as [5], the proposed architecture enables fully on-chip implementation without external memory.

Overall, the proposed design achieves a favorable trade-off among resolution, computational complexity, and memory usage, making it well suited for real-time 3-D ultrasonic imaging on resource-constrained platforms.

Table 10: Comparison of 3D ultrasonic imaging systems

	This Work	JSSC'22 [23]	IEEE Trans [5]
Spatial Resolution (°)	2.3	2.7	12
FPS	68.56	24	29
Platform	FPGA (ZCU102)	ASIC (180 nm)	FPGA + GPU
Voxel Number	$69 \times 69 \times 284$	$64 \times 64 \times 100$	$51 \times 51 \times 3500$
Range (m)	0.3–2	7	0.017–3

Chapter 6. Conclusions

This thesis proposes a depth-sparse 3-D synthetic aperture ultrasonic imaging system to enable further exploration of ultrasonic sensing applications. The proposed architecture incorporates multi-level pipelining and parallel processing to handle the high computational demand of beamforming. In addition, the complexity of delay calculation is reduced by applying a linear approximation method.

To overcome the limited resolution of conventional ultrasonic imaging systems, a synthetic aperture (SA) algorithm is adopted, which effectively improves the angular resolution as demonstrated in Chapter 5.

Furthermore, to address the high memory requirement of synthetic aperture imaging, this thesis proposes a depth-sparse memory architecture that stores only valid voxels along the depth dimension. This strategy significantly reduces memory usage while eliminating frequent communication with external memory. As a result, the processing pipeline can operate continuously without being limited by off-chip memory latency.

Future work will focus on integrating the proposed imaging system with the ultrasonic front-end hardware, including the transmitter and receiver array modules. This integration will enable the complete end-to-end validation of the system in a practical ultrasonic sensing environment.

The current implementation targets the ZCU102 FPGA platform with a clock period of 8 ns. With the rapid development of next-generation FPGA devices and emerging components such as AI Engines, further performance improvements can be achieved by exploring additional design space on these advanced architectures.

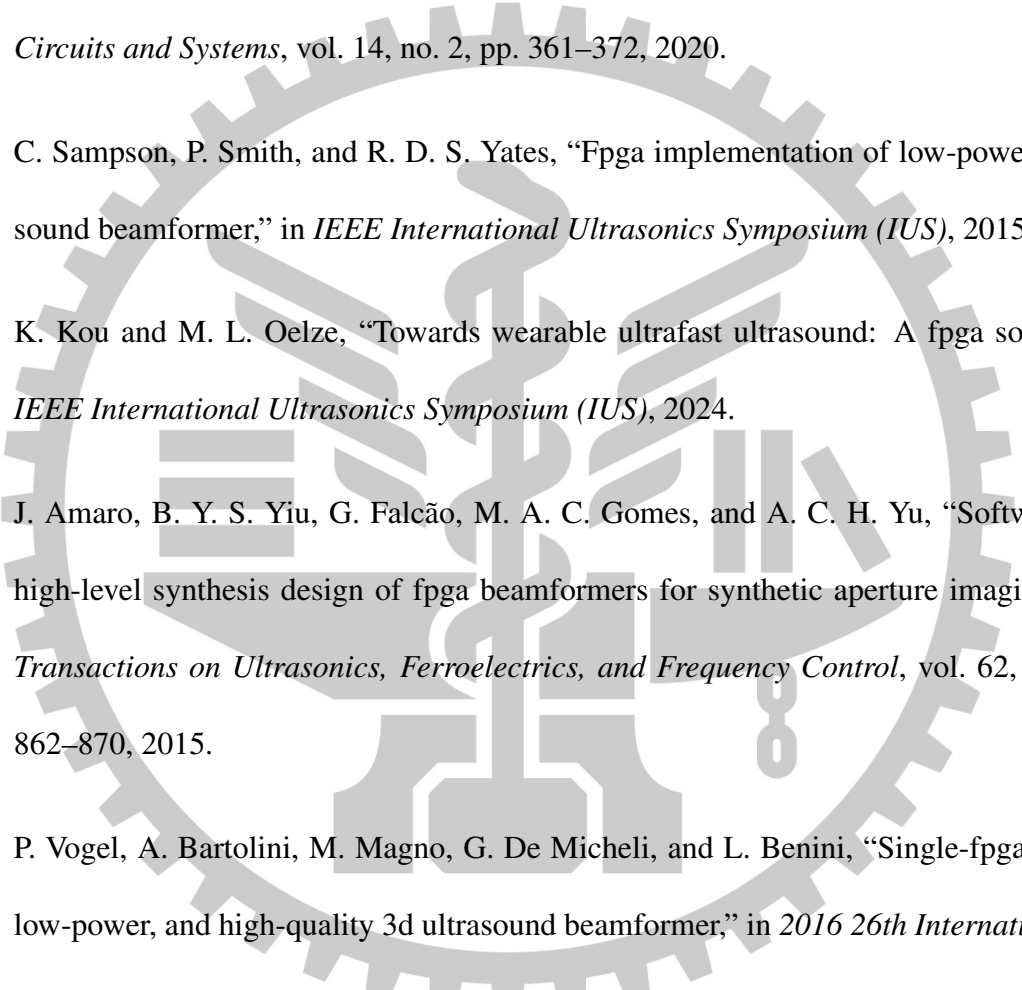
Finally, the proposed architecture is not limited to FPGA implementation. Due to its streaming dataflow structure and hardware-oriented design, the architecture can also be migrated to an ASIC platform, which could further improve performance and energy efficiency.



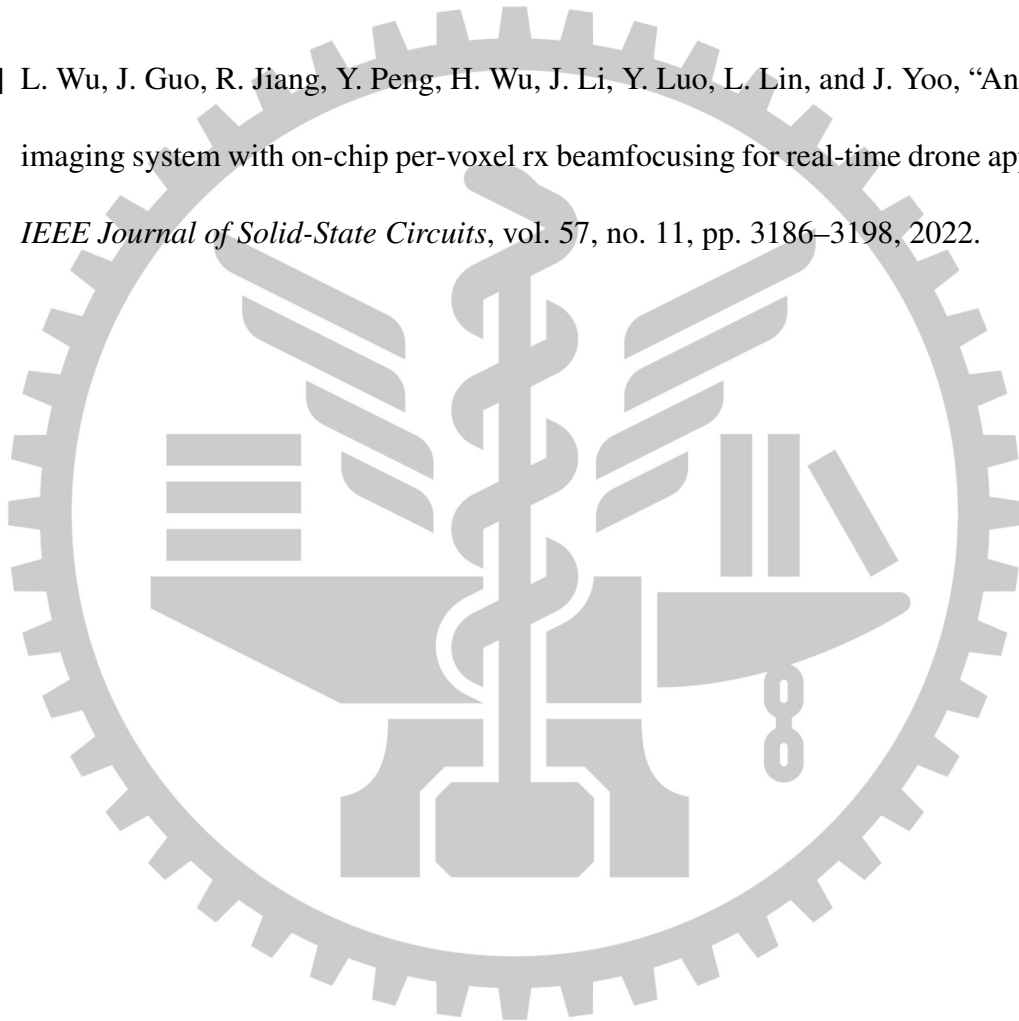
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Appendix 附錄

